Open-book reference: EDC Weeks 1–10 (full course)

Cheat sheet · Use from Wed 18 Nov · pack REFE

Covers every week, 1 to 10, and replaces the short Weeks 1–5 strip of the earlier version. Boylestad notation throughout (r_e model, hybrid-π, I_{DSS}, V_P, k). **Work in mA, kΩ, mS and V**: then mA × kΩ = V and mS × kΩ is a plain gain. $V_T = kT_K/q$ is the thermal voltage, 25.875 mV ≈ 26 mV at 27 °C. BJT small-signal work uses $r_e = 26\ \text{mV}/I_E$ (WS6 Q1–Q3 use 25 mV: follow the question). For FETs, $V_{GS(\text{Th})}$ is the E-MOSFET threshold (written V_T in the Sedra questions; it is not the thermal voltage). r_o or r_d not given means ∞: say so in your answer. Course constants: $V_{BE,on} = 0.7$ V, $V_{CE,sat} = 0.2$ V, Si knee 0.7 V. All check values below were recomputed in Python.

What the test formula table prints (practice test p. 2): $R = \rho L/A$; $\sigma = q(n\mu_n + p\mu_p)$; $v_d = \mu E$, $E = V/L$; $n_0 p_0 = n_i^2$; $n_i = BT^{3/2}e^{-E_g/2kT}$; $V_T = kT_K/q$, $k = 1.38 \times 10^{-23}$ J/K, $q = 1.6 \times 10^{-19}$ C; Shockley; Zener TC; half-wave i_{Dav}, i_{Dmax} and $V_r = V_P/(fCR)$; knee voltages; %V.R.; BJT $n_p(0)$, I_n, $i_C = I_S e^{v_{BE}/V_T}$, I_S, I_{CEO}, α, β. **Not printed:** B, E_g, the full-wave ripple ($2f$), PIVs, V_{dc}, every bias and small-signal formula. Those are on this sheet.

Weeks 1–2 Semiconductors and diodes

Circuit tools (WS1) and the ideal op-amp

Tool	Rule	Trap
Voltage divider	$V_i = V_{in}R_i/\sum R$	series elements only. A load on the middle node breaks it
Current divider (2 R)	$I_1 = I\,\dfrac{R_2}{R_1 + R_2}$	the *other* resistor goes on top
Superposition	one independent source at a time; V source off = short, I source off = open	dependent sources stay ON
Thévenin	V_{Th} = open-circuit voltage (load removed); R_{Th} = resistance with independent sources off. With dependent sources: $R_{Th} = V_{oc}/I_{sc}$; only dependent sources: V_{test}/I_{test}	forgetting to remove the load before finding V_{oc}
Norton	$I_N = V_{Th}/R_{Th}$ (short-circuit current), $R_N = R_{Th}$	arrow head points to the + of V_{Th}
Nodal	reference node, $N-1$ KCLs in node voltages. V source between two unknown nodes: supernode + $V_a - V_b = V_s$	draw current arrows first
Bias node X fed by (V_k, R_k)	$V_{Th} = \dfrac{\sum_k V_k/R_k}{\sum_k 1/R_k}$, $R_{Th} = R_1 \parallel R_2 \parallel \cdots$ (a resistor to ground is a 0 V branch)	V_{Th} must lie between the most negative and most positive supply
Standard base divider	$V_{Th} = E_{Th} = \dfrac{R_2 V_{CC}}{R_1 + R_2}$, $R_{Th} = R_1 \parallel R_2$	two supplies: superposition, never treat $-V_{EE}$ as ground
Power sign	passive convention: current into +, $p = vi > 0$ absorbs	sources usually supply ($p < 0$ passive)
DC steady state	capacitor = open, inductor = short	
Ideal op-amp	$i_+ = i_- = 0$; with negative feedback $v_+ = v_-$; KCL at the − node, never at the output; $\lvert v_o\rvert$ limited by the rails	inverting $-R_f/R_1$, non-inverting $1 + R_f/R_1$

CHECK VALUES. PT Q7: +9 V via 15 kΩ and −12 V via 30 kΩ gives $V_{Th} = 9(30/45) + (-12)(15/45) = 2$ V, $R_{Th} = 10$ kΩ. WS5 Q15 collector node (+5 V via 3 kΩ, −10 V via 27 kΩ): 3.5 V, 2.7 kΩ. WS5 Q14 base (20 V, 90 k/10 k): 2 V, 9 kΩ. WS1 Q24 nodal: $V_C = 0.2$ V. WS4 Q2: $v_o = -10v_i + 4$ with ±12 V rails stays linear for $-0.8 \le v_i \le 1.6$ V.

Semiconductor physics (Week 1 Part 2, WS2)

Quantity	Formula	Trap
Intrinsic concentration	$n_i = BT^{3/2}e^{-E_g/(2kT)}$, Si: $B = 7.3 \times 10^{15}$ $\text{cm}^{-3}\text{K}^{-3/2}$, $E_g = 1.12$ eV, $k = 8.62 \times 10^{-5}$ eV/K	k in eV/K here (not J/K); T in kelvin. Formula gives 1.49×10^{10} at 300 K; use the quoted 1.5×10^{10}
Mass action	$n_0 p_0 = n_i^2$ (thermal equilibrium, always)	not $n + p = n_i$
Intrinsic	$n = p = n_i$, $\sigma = q n_i(\mu_n + \mu_p)$	
n-type (donors, group V: P, As, Sb)	$n_n \approx N_D$, $p_n = n_i^2/N_D$	valid if $N_D \geq 10 n_i$
p-type (acceptors, group III: B, Al, Ga, In)	$p_p \approx N_A$, $n_p = n_i^2/N_A$	doped material is still electrically **neutral**
Exact (n-type)	$n = \frac{N_D}{2} + \sqrt{\left(\frac{N_D}{2}\right)^2 + n_i^2}$, $p = n_i^2/n$	use when $n_i \gtrsim 0.1 N_D$ (hot, light doping). p-type: swap $N_D \to N_A$, $n \leftrightarrow p$
Conductivity	$\sigma = q(n\mu_n + p\mu_p)$ $(\Omega\text{-cm})^{-1}$, $\rho = 1/\sigma$	minority term negligible when doped: say so
Resistance	$R = \rho L/A$	cm and cm²: 1 µm = 10^{-4} cm, 1 µm² = 10^{-8} cm², 1 mm = 0.1 cm
Drift	$v_d = \mu E$, $E = V/L$, $J = \sigma E$	electrons drift against E; their current is along E
Diffusion	$J_p = -qD_p\, dp/dx$, $J_n = +qD_n\, dn/dx$	high to low concentration
Einstein	$D_n/\mu_n = D_p/\mu_p = V_T$	
Thermal voltage	$V_T = kT_K/q$; 27 °C: 25.875 mV; 25 °C: 25.72 mV; 20 °C: 25.27 mV	$T_K = 273 + T(^\circ\text{C})$
Mobilities	WS2: $\mu_n = 1350$, $\mu_p = 500$ cm²/V·s; lecture slide 20: 1300/500	use what is given

pn junction (qualitative, MCQs). Open circuit: majority carriers diffuse and recombine, leaving immobile ionised dopants (+ on n side, − on p side) = depletion region; its field sets the barrier V_0 and drives the minority drift current I_S; at equilibrium $I_D = I_S$ (no net current). Charge balance $x_n N_D = x_p N_A$, so $x_n/x_p = N_A/N_D$: the region extends into the **lightly doped** side; $W \approx 0.1$–1 µm. $V_0 = V_T \ln(N_A N_D/n_i^2)$. Forward bias: barrier lowered, W shrinks, current rises exponentially. Reverse bias: W grows, only I_S (nA) flows, independent of V but doubling every 10 °C. Breakdown beyond PIV/PRV (Zener or avalanche). Junction capacitance C_j falls as V_R rises (varactor). Reverse recovery $t_{rr} = t_s + t_t$ (ns to 1 µs).

> **CHECK VALUES (WS2, RECOMPUTED).** Cylinder L = 0.8 cm, D = 0.1 cm ($A = 7.85 \times 10^{-3}$ cm²): intrinsic Si $\sigma = 4.44 \times 10^{-6}$, $\rho = 2.25 \times 10^5$ Ω-cm, $R = 22.9$ MΩ; copper ($\sigma = 5.65 \times 10^5$) 0.18 mΩ. 5 V across it: $E = 6.25$ V/cm, $v_{dn} = 8.44 \times 10^3$ cm/s, $v_{dp} = 3.13 \times 10^3$ cm/s. $N_D = 10^{14}$: $p = 2.25 \times 10^6$ cm^{-3}, $\sigma = 0.0216$. 50 °C intrinsic: $n_i = 7.88 \times 10^{10}$, $\rho = 4.29 \times 10^4$ Ω-cm; bar 5 µm × (3 µm × 3 µm): $R = 238$ MΩ (1.25 GΩ at 300 K, an 81 % drop); doped $N_D = 5 \times 10^{15}$: $\rho = 0.926$ Ω-cm, $R = 5.14$ kΩ. WS2 Q5 ($N_A = 2 \times 10^{15}$): $n = 1.125 \times 10^5$ at 300 K, 5×10^{10} at 405 K ($n_i = 10^{13}$). V_0 for $N_A = 10^{17}$, $N_D = 10^{15}$: 0.694 V.

The diode equation, resistance levels and temperature

Formula	Use	Trap
$I_D = I_S\left(e^{V_D/nV_T} - 1\right)$; forward $I_D \approx I_S e^{V_D/nV_T}$; reverse $I_D \approx -I_S$	Shockley; n = ideality factor, 1 to 2	the -1 matters only near $V_D \approx 0$
$V_D = nV_T \ln(I_D/I_S)$	one point, find V_D	keep units equal inside the log; do not drop n
$\Delta V_D = nV_T \ln(I_2/I_1)$	change in current: I_S cancels	**ln, not** $\log_{10}$. One decade = $2.303\,nV_T$: 59.6 mV ($n=1$), 89.4 mV ($n=1.5$) at 27 °C
$n = \dfrac{V_2 - V_1}{V_T \ln(I_2/I_1)}$, then $I_S = I_1 e^{-V_1/nV_T}$	two (I, V) points	check I_S with the second point
Two identical diodes in parallel on one current source	each carries $I/2$: V falls by $nV_T \ln 2 \approx 18$ mV	
$R_D = V_D/I_D$	DC (static) resistance at one point	lecture: 250 Ω at 2 mA, 40 Ω at 20 mA, 10 MΩ at −10 V
$r_d = \Delta V_d/\Delta I_d$ (tangent) $= nV_T/I_D$	AC / dynamic / small-signal, valid for $v_d \lesssim 5$ mV	Week 2 writes V_T/I_D ($n=1$): 26 mV/10 mA = 2.6 Ω
$r_{av} = \Delta V_d/\Delta I_d$ between two stated points	builds the 3rd model	point to point, not a tangent
Temperature	forward curve shifts **left** 2.5 mV/°C at fixed I; $I_S(T_2) = I_S(T_1)\,2^{\Delta T/10}$; $V_T \propto T_K$	hotter: smaller V_D, larger leakage
Load line (series V_{DD}, R)	$I_D = (V_{DD} - V_D)/R$; Q where it meets the diode curve	intercepts V_{DD} and V_{DD}/R

The three piecewise-linear models, exactly as the course numbers them (WS2 Q17, WS3 Q15, PT Q3):

Model	Course name	ON (conducting)	OFF	ON valid if	OFF valid if
1st	ideal model	short circuit, $v_d = 0$	open circuit	$i_d > 0$	$v_d < 0$
2nd	voltage drop model (constant drop, Boylestad "simplified"): Si 0.7 V unless given	ideal diode + V_D source (+ on the anode side)	open circuit	$i_d > 0$	$v_d < V_D$
3rd	voltage drop with a resistor model (Boylestad "piecewise-linear": V_K + r_{av})	ideal diode + V_D source + r_D in series; $v_d = V_D + i_d r_D$	open circuit	$i_d > 0$	$v_d < V_D$

Reverse-biased diodes are an open circuit in **every** model. The real curve has four regions (breakdown, reverse, partial conduction, full conduction), so full linearisation needs four segments (WS2 Q14–15). Knee V_K: Ge 0.3 V, Si 0.7 V, GaAs 1.2 V. Shortcut for one diode: Thévenin everything else, then $i_d = (V_{Th} - V_D)/(R_{Th} + r_D)$.

> **CHECK VALUES.** PT Q2 ($n = 1.5$, 27 °C, 0.1 → 10 mA): $\Delta V_D = 1.5(25.875)(4.605) =$ **0.179 V**. WS2 Q8 ($I_S = 0.2$ pA, $V_T = 25$ mV): 0.558, 0.616, 0.673 V at 1, 10, 100 mA. WS2 Q9 (1 mA at 0.7 V, 10 mA at 0.8 V, 25 mV): $n = 1.737$, $I_S = 0.1$ nA. WS3 Q1 (25 °C, 1 µA, 100 mA): 296 mV. WS3 Q2 ($I_S = 1$ µA at 20 °C into 1 MΩ): 4 V at 40 °C, 0.25 V at 0 °C. WS3 Q4 (20 °C, $n = 2$, 40 nA, 0.5 V): 0.79 mA. WS3 Q3 (three diodes give 2.4 V from 10 V, +0.1 V/decade from 0.7 V at 1 mA): 10 mA, $R = 760$ Ω. WS3 Q5: $r_{av} = 0.12/26\text{ mA} = 4.62$ Ω (Ge 0.3 V, GaAs 1.2 V). WS3 Q6: $R_D = 175$ Ω at 4 mA. WS3 Q15 (10 V, 50 Ω, 5 Ω across the diode): ideal 0.2 A; 2nd 46 mA; 3rd ($r_D = 10$ Ω) 14.4 mA, $v_d = 0.844$ V. WS3 Q16: $V_i = +10$: D1 ON 3.19 mA, $v_{d2} = -6.81$ V; $V_i = -10$: D2 ON 5.81 mA, $v_{d1} = -5.19$ V. WS3 Q17 OR gate (0.5 V, 1 k, 9 k): one input high 4.05 V; both high 4.26 V (0.237 mA each). **PT Q3** (Thévenin at A = 5 V, 2.5 kΩ; three diodes 0.6 V + 0.2 kΩ; 10 kΩ to +3 V): $I = 0.2/13.1 =$ **15.3 µA** > 0 so all conduct, $V_o =$ **3.15 V**.

Rectifiers and the capacitor filter

$V_m = \sqrt{2}\,V_{s(rms)}$. Subtract only the drops in the conduction path.

	Half-wave	Centre-tap full-wave	Bridge full-wave
Diodes / drop in path	1 / V_D	2 / V_D	4 / $2V_D$
Peak out V_P	$V_m - V_D$	$V_m - V_D$ (V_m of one half-winding)	$V_m - 2V_D$
V_{dc}, no C	$V_P/\pi = 0.318V_P$	$2V_P/\pi = 0.636V_P$	$2V_P/\pi$
V_{rms} (ideal diodes)	$V_m/2$	$V_m/\sqrt{2}$	$V_m/\sqrt{2}$
Output (ripple) frequency	f	$2f$	$2f$
PIV	V_S (peak); $\approx 2V_m$ **with a filter C**	$2V_S - V_D$	$V_S - V_D$
η_{max}	40.6 %	81.2 %	81.2 %

Choose a diode rated $\geq 1.5\times$ PIV. **With diode resistance** r_f (WS3 Q20–21): $I_m = V_m/(r_f + R)$ (bridge $2r_f + R$); HW $I_{dc} = I_m/\pi$, $I_{rms} = I_m/2$; FW $I_{dc} = 2I_m/\pi$, $I_{rms} = I_m/\sqrt{2}$; $P_{ac} = I_{rms}^2(r_f + R)$, $P_{dc} = I_{dc}^2 R$, $\eta = P_{dc}/P_{ac}$; each bridge diode conducts half the cycle, $I_{rms,D} = I_m/2$.

Filter quantity ($CR \gg T$)	Half-wave	Full-wave
Ripple V_r (peak-to-peak)	$\dfrac{V_P}{fCR} = \dfrac{I_L}{fC}$ (on the formula sheet)	$\dfrac{V_P}{2fCR} = \dfrac{I_L}{2fC}$ (not on the sheet)
Choose C	$C = \dfrac{V_P}{fRV_r}$	$C = \dfrac{V_P}{2fRV_r}$
DC output, load current	$V_{O(dc)} = V_P - \frac{1}{2}V_r$, $I_L = V_{O(dc)}/R \approx V_P/R$	same
Conduction angle	$\omega\Delta t \approx \sqrt{2V_r/V_P}$ rad; fraction $= \omega\Delta t/2\pi$; $\Delta t = \omega\Delta t/(2\pi f)$	same
Average diode current during conduction	$i_{Dav} = I_L\left(1 + \pi\sqrt{2V_P/V_r}\right)$	$I_L\left(1 + \pi\sqrt{V_P/2V_r}\right)$ (Sedra; say which you used)
Peak diode current	$i_{Dmax} = I_L\left(1 + 2\pi\sqrt{2V_P/V_r}\right)$	$I_L\left(1 + 2\pi\sqrt{V_P/2V_r}\right)$
Ripple factor	$r = V_{r(rms)}/V_{dc}$, $V_{r(rms)} = V_{r(pp)}/(2\sqrt{3})$: 10 % means $V_{r(pp)} = 0.346V_{dc}$	same
Regulation	$\%\text{V.R.} = \dfrac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$ (0 % ideal)	divide by **full load**

Supply chain: transformer → rectifier → filter → regulator. Ripple derivation: $V_P e^{-T/CR} \approx V_P(1 - T/CR)$, so $V_r \approx V_P T/CR$; full-wave discharges for $T/2$.

> **CHECK VALUES.** WS3 Q9 (12 V rms, 50 Hz, $V_D = 0.8$, 100 Ω, $V_r \leq 1$ V, half-wave): $V_m = 16.97$, $V_P = 16.17$ V, $C = 3.23$ mF, $V_{O(dc)} = 15.67$ V, $I_L = 157$ mA, PIV $= 2V_m = 33.9$ V. WS3 Q14 ($V_P = 100$ V, 60 Hz, 10 kΩ, $V_r = 2$ V): $C = 83.3$ µF, $I_L = 10$ mA, $\omega\Delta t = 0.2$ rad (3.18 %, $\Delta t = 0.53$ ms), $i_{Dav} = 324$ mA, $i_{Dmax} = 638$ mA. WS3 Q20 (HW, $50\sin\omega t$, $r_f = 20$, $R = 800$): $I_m = 61.0$ mA, $I_{dc} = 19.4$ mA, $I_{rms} = 30.5$ mA, $V_{dc,L} = 15.5$ V, $\eta = 39.5$ %. WS3 Q21 (bridge, $240\sin 314t$, $r_f = 20$ each, 460 Ω): $I_m = 0.48$ A, $I_{dc} = 0.306$ A, $V_{dc} = 140.6$ V, PIV 240 V, $f_o = 100$ Hz, 1.15 W per diode. WS3 Q11 (full-wave-shaped 50 V peak): $V_{dc} = 31.8$ V. Bridge 24 V rms, 50 Hz, 500 Ω, 470 µF, 0.7 V: $V_P = 32.5$ V, $V_r = 1.38$ V, $V_{O(dc)} = 31.8$ V, PIV 33.2 V.

Zener shunt regulator

Circuit: V_s → R_S → output node; Zener (cathode up, reverse breakdown, conducts against its arrow) and R_L from the node to ground.

Item	Formula
Zener model	$V_Z = V_{Z0} + r_z I_Z$, $\Delta V = r_z \Delta I$; ideal: $r_z = 0$
KVL, KCL	$I_S = \dfrac{V_s - V_O}{R_S}$, $I_S = I_Z + I_L$, $I_L = V_O / R_L$
Analysis with r_z	$\dfrac{V_O - V_{Z0}}{r_z} + \dfrac{V_O}{R_L} = \dfrac{V_s - V_O}{R_S}$, solve for V_O; $P_Z = V_O I_Z$
Ideal Zener	$V_O = V_Z$; I_S fixed by V_s and R_S only; the load just shares it ($I_Z = I_S - I_L$)
Is it on?	$I_Z > 0$ (or $\geq I_{ZK}$); if not, $V_O = V_s R_L / (R_L + R_S) < V_Z$
Design R_S (lowest supply, heaviest load)	$R_S = \dfrac{V_{s,min} - V_Z}{I_{L,max} + I_{Z,min}}$, $I_{Z,min} \approx 0$ if not given
Zener rating (highest supply, lightest load)	$I_{Z,max} = \dfrac{V_{s,max} - V_Z}{R_S} - I_{L,min}$, $P_{Z,max} = V_Z I_{Z,max}$
Load range for a given Zener	$I_{L,max} = I_{R,min} - I_{Z,min}$, $R_{L,min} = V_Z / I_{L,max}$; $I_{L,min} = I_{R,max} - I_{Z,max}$, $R_{L,max} = V_Z / I_{L,min}$ (∞ if < 0)
Filtered input	use the ripple valley $V_P - V_r$ as $V_{s,min}$
Line / load regulation (textbook)	$\Delta V_O / \Delta V_s = \dfrac{r_z \parallel R_L}{R_S + r_z \parallel R_L}$; $\Delta V_O / \Delta I_L = -(r_z \parallel R_S)$
Temperature coefficient (formula sheet)	$TC = \dfrac{\Delta V_Z / V_Z}{T_1 - T_0} \times 100\ \%/^\circ\mathrm{C}$

CHECK VALUES. WS3 Q10 ($V_{ZK} = 10$, $r_z = 20\ \Omega$, 15 V, $R_S = 100$, $R_L = 500$): $V_O = 10.48$ V, $I_Z = 24.2$ mA, $I_L = 21.0$ mA, $I_S = 45.2$ mA, $P_Z = 254$ mW; line regulation 0.161 V/V, load regulation −16.7 Ω. **PT Q4 = WS3 Q13** (16 V, $V_L = 12$ V, $0 \leq I_L \leq 200$ mA): $V_Z = 12$ V, $R_S = 4/0.2 = \mathbf{20\ \Omega}$; at no load $I_Z = 200$ mA, $P_Z = 2.4$ W. WS3 Q22 ($V_Z = 4.8$, $0.2 \leq I_Z \leq 7$ mA, 1 kΩ, 10 ± 1 V): $R_{L,min} = 1.2$ kΩ, $R_{L,max} = \infty$. Ideal 6.2 V Zener, 12 V, 220 Ω, 1 kΩ: $I_S = 26.4$, $I_L = 6.2$, $I_Z = 20.2$ mA, $P_Z = 125$ mW.

Clippers, clampers, the voltage doubler and special diodes

Circuit	Rule
Any clipper	assume the diode OFF, write $v_d(v_i)$, set $v_d = V_D$ (0 if ideal) to get the transition voltage v_{iT}, write one linear v_o per region, sketch v_o under v_i on the same axes
Series clipper	diode in the signal path; OFF gives $v_o = 0$ (no current in the load); a battery shifts the transition to $v_i = \pm V_B \pm V_D$ (signs from the polarities)
Parallel clipper	diode (+ battery) across the output; ON holds v_o at $V_B \pm V_D$; OFF v_o follows v_i (or a divider of it)
Clamper (series C, shunt diode)	C charges to the input peak (minus V_D) through the diode, then holds it; the whole waveform shifts by that level, same shape and same peak-to-peak. Diode cathode to ground, anode at output: output between 0 and $-2V_m$ (reverse it for 0 to $+2V_m$)
Voltage doubler	clamper + peak detector: $C_1 \to V_m$, D_2 sees $v_i + V_m$, $C_2 \to 2V_m$ (minus drops); each diode blocks $2V_m$
Output shape for V_{dc}	half-wave-shaped: $0.318 V_{op}$; full-wave-shaped: $0.636 V_{op}$

Special diode	Bias	Key fact
LED	forward	$E = hf = hc/\lambda \approx E_g$: $\lambda(\text{nm}) \approx 1240/E_g(\text{eV})$; eye 350–800 nm, peak 550 nm; drop about 1.5–3 V
Schottky	forward	metal–n junction, low V_F (0.3–0.5 V), fast (no minority storage)
Varactor	reverse	C_j falls as V_R rises (depletion widens); tuning
Photodiode	reverse	photocurrent (reverse) proportional to light intensity
Zener	reverse breakdown	conducts against the arrow at V_Z; model $V_{Z0} + r_z I_Z$

CHECK VALUES. **PT Q5 = WS3 Q12** (square wave +20 V / −5 V, 4 V battery opposing, ideal diode, 1 kΩ): $v_{iT} = -4$ V; $v_o =$ **0 V** for +20 V, **−1 V** for −5 V. WS3 Q18 (diode 0.7 V across the 1 kΩ series resistor, output across 0.5 kΩ; this reading matches the solution's equations): $v_{iT} = 1.05$ V; $v_o = v_i/3$ below, $v_o = v_i - 0.7$ above (both 0.35 V at the transition).

Weeks 3–5 BJTs

Physics, modes, α and β

npn = n emitter (heavily doped) / p base (**thin, lightly doped**) / n collector. In active mode the forward EBJ injects electrons, about 1 % recombine in the thin base, the rest diffuse across and are swept into the collector by the reverse CBJ (diffusion dominates at the EBJ, drift at the CBJ).
$i_C = I_S e^{v_{BE}/V_T}$, $I_S = \dfrac{A_E q D_n n_i^2}{N_A W}$, $n_p(0) = n_{p0} e^{v_{BE}/V_T}$,
$I_n = A_E q D_n \dfrac{dn_p}{dx} = -A_E q D_n \dfrac{n_p(0)}{W}$. npn: I_B, I_C in, I_E out; pnp all reversed. The emitter arrow points along conventional current. PT Q1 IV (a), Q1 V (b).

Mode	EBJ	CBJ	DC model	Check
Active	forward	reverse	$V_{BE} = 0.7$ V source B–E, βI_B source C→E	$I_B > 0$ and $V_{CE} > 0.2$ V
Saturation	forward	forward	$V_{BE} = 0.7$ V, $V_{CE} = 0.2$ V source (closed switch)	$I_B > 0$, $I_C > 0$, $I_C < \beta I_B$ (forced $\beta = I_C/I_B < \beta$)
Cut-off	reverse	reverse	open circuit, $I_B = I_C = I_E = 0$	$V_{BE} < 0.7$ V

The **WS4 handout table has the saturation and cut-off rows swapped**; the lecture table above is correct (WS4's own Q6.2 answer agrees). From node voltages (npn): active if $V_{BE} = 0.7$ and $V_{BC} < 0$ ($V_C > V_B$).

Relation	Formula	Trap
KCL (all modes)	$I_E = I_C + I_B$	two currents given: subtract first
Active	$I_C = \beta I_B = \alpha I_E$, $I_E = (\beta+1) I_B$	not in saturation
Conversion (formula sheet)	$\alpha = \dfrac{\beta}{\beta+1}$, $\beta = \dfrac{\alpha}{1-\alpha}$	0.98 → 49, 0.99 → 99, 0.999 → 999: keep 4 s.f. on α
Leakage	$I_C = \alpha I_E + I_{CBO}$, $I_{CEO} = \dfrac{I_{CBO}}{1-\alpha} = (\beta+1) I_{CBO}$	I_{CEO} is the $I_B = 0$ curve
Gains from curves	$\beta_{dc} = I_{CQ}/I_{BQ}$ ("β at the Q-point"); $\beta_{ac} = \Delta I_C / \Delta I_B$ at fixed V_{CE}	use the Q values themselves for β_{dc}
Early effect	$i_C = I_S e^{v_{BE}/V_T}(1 + v_{CE}/V_A)$, $r_o = \dfrac{V_A + V_{CE}}{I_C} \approx \dfrac{V_A}{I_C}$	curves meet at $v_{CE} = -V_A$; ignore for DC unless V_A given
Ratings	$P_C = V_{CE} I_C \le P_{C,max}$	

Configurations by common terminal: CB (input I_E–V_{BE}, output I_C–V_{CB}, $I_C \approx I_E$), CE (output I_C–V_{CE} with I_B steps: draw load lines here; saturation near 0.2 V, cut-off below $I_B = 0$), CC (emitter follower, output at the emitter).

> **CHECK VALUES.** WS4 Q9: $I_E = 5$ mA, $I_B = 0.1$ mA gives $I_C = 4.9$ mA, α = 0.98, β = 49; $I_C = 8$, $I_B = 0.1$ gives β = 80, α = 0.988. pnp β = 50, $I_C = 4$ mA: $I_B = 0.08$, $I_E = 4.08$ mA. WS4 Q10 (β = 50): 100 µA/5 mA active, 100 µA/1 mA saturated, 0/0 cut-off. WS5 Q3: $I_E = 0.4$ mA, $I_B = 6$ µA, $I_C = 0.394$ mA, α = 0.985, β = 65.7.

DC bias configurations

Every circuit is two loops: base–emitter (sets I_B) and collector–emitter (sets V_{CE}). R_E seen from the base loop is $(\beta+1)R_E$ (Lecture 4: $R_i = (\beta+1)R_E$).

Configuration	Base loop (active)	Output loop	$I_{C,sat}$
Fixed bias (R_B from V_{CC}, emitter grounded)	$I_B = \dfrac{V_{CC} - V_{BE}}{R_B}$	$V_{CE} = V_{CC} - I_C R_C = V_C$	V_{CC}/R_C
Emitter-stabilised (R_E added)	$I_B = \dfrac{V_{CC} - V_{BE}}{R_B + (\beta+1)R_E}$	$V_{CE} = V_{CC} - I_C R_C - I_E R_E$	$\dfrac{V_{CC}}{R_C + R_E}$
Voltage divider, **exact**	$R_{Th} = R_1 \parallel R_2$, $E_{Th} = \dfrac{R_2 V_{CC}}{R_1 + R_2}$, $I_B = \dfrac{E_{Th} - V_{BE}}{R_{Th} + (\beta+1)R_E}$	as above	$\dfrac{V_{CC}}{R_C + R_E}$
Voltage divider, **approximate**, only if $\beta R_E \geq 10 R_2$ (write the test)	$V_B = E_{Th}$, $V_E = V_B - 0.7$, $I_C \approx I_E = V_E/R_E$, $I_B = I_C/\beta$	$V_{CE} \approx V_{CC} - I_C(R_C + R_E)$	same
Collector feedback (R_F C to B)	slides: $I_B = \dfrac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$; exact $(\beta+1)$ in place of β (R_C carries I_E)	$V_{CE} = V_{CC} - I_E(R_C + R_E)$	
Emitter follower (C at rail, E via R_E to $-V_{EE}$, B via R_B to ground)	$I_B = \dfrac{V_{EE} - V_{BE}}{R_B + (\beta+1)R_E}$	$V_{CE} = V_{EE} - I_E R_E$	
Common base (E via R_E to $-V_{EE}$, B grounded, C via R_C to V_{CC})	$I_E = \dfrac{V_{EE} - V_{BE}}{R_E}$, $I_B = I_E/(\beta+1)$	$V_{CB} = V_{CC} - I_C R_C$, $V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$	
General single BJT	$I_B = \dfrac{V_{Th} - V_{BE}}{R_{Th} + (\beta+1)R_E}$	node voltages: $V_E = I_E R_E$, $V_C = V_{CC} - I_C R_C$, $V_B = V_E + 0.7$, $V_{BC} = V_B - V_C$	

BJT DC bias configurations: fixed, emitter-stabilised, voltage divider, collector feedback, emitter follower, common base

Why the divider is used: the approximate $I_C = (V_B - 0.7)/R_E$ contains no β, so the Q-point hardly moves with β. **Load line:** $V_{CE} = V_{CC} - I_C(R_C + R_E)$, intercepts $(V_{CC}, 0)$ and $(0, V_{CC}/(R_C + R_E))$; Q where it meets the I_{BQ} curve (interpolate between printed curves). Raising R_C pivots the line about V_{CC}; lowering V_{CC} shifts it parallel; raising I_B slides Q towards saturation. **Design from a load line:** V_{CC} = V_{CE} intercept, $R_C(+R_E) = V_{CC}/I_{C,sat}$, $R_B = (V_{CC} - 0.7 - I_E R_E)/I_{BQ}$, then nearest standard value and % error.

> **CHECK VALUES.** **PT Q6** (20 V, $R_B = 270$ k, $R_C = 470$ Ω, $R_E = 2.2$ k, β = 125): $I_{C,sat} = 7.49$ mA, $I_{BQ} = 19.3/547.2\text{k} = 35.3$ µA; analytical $I_C = 4.41$ mA, $V_{CE} = 8.15$ V; graph reading about 4.6 mA, 7.6 V, so $\beta_{dc} \approx 130$ (the Test 1 Reference's 4.2 mA reading is too low; see EPT). **PT Q7** (Thévenin 2 V, 10 kΩ, emitter grounded, β = 50): $I_B = 130$ µA, $I_C = 6.5$ mA, $V_C = 2.25$ V > 0.2, active. WS5 Q7 (exact: 16 V, 62 k/9.1 k, 3.9 k, 0.68 k, β = 80; $\beta R_E = 54.4$ k < 91 k): $R_{Th} = 7.94$ k, $E_{Th} = 2.048$ V, $I_B = 21.4$ µA, $I_C = 1.71$ mA, $V_{CE} = 8.15$ V, $V_E = 1.18$, $V_C = 9.33$, $V_B = 1.88$ V (approximate would give 1.98 mA, 6.92 V: 16 % off). WS5 Q8 (approx valid, 18 V, 39 k/8.2 k, 3.3 k, 1 k, β = 120): $V_B = 3.13$ V, $I_E = 2.43$ mA, $I_B = 20.2$ µA, $V_{CE} = 7.56$ V. WS5 Q9: $I_B = 0.174$ mA, $V_{CE} = 2.44$ V, active. WS5 Q14: $I_B = 21.7$ µA, $I_C = 1.08$ mA, $V_{CE} = 13.5$ V. WS5 Q15: $V_o = 1.10$ V. Lecture: collector feedback (10 V, 250 k, 4.7 k, 1.2 k, β = 90) $I_C = 1.07$ mA, $V_{CE} = 3.68$ V (exact 1.06 mA, 3.65 V); emitter follower (20 V, 240 k, 2 k, β = 90) $I_B = 45.7$ µA, $I_E = 4.16$ mA, $V_{CE} = 11.68$ V; CB (4 V, 1.2 k, 10 V, 2.4 k, β = 60) $I_E = 2.75$ mA, $V_{CB} = 3.51$ V, $V_{CE} = 4.1$ V (slide 46 prints "24 kΩ": it means 2.4 kΩ). Load-line design (20 V, 8 mA, 40 µA): $R_C = 2.5 \to 2.4$ kΩ, $R_B = 482.5 \to 470$ kΩ (41.1 µA).

Operating state, pnp, β range, design and two-BJT DC

Topic	Rule
Failed active check	$V_{CE} \leq 0.2$ (often negative) → saturation; $I_B \leq 0$ or $I_E < 0$ → cut-off. A negative V_{CE} is not an arithmetic slip
Saturation solve	$V_{BE} = 0.7$, $V_{CE} = 0.2$; unknowns I_B, I_C; base loop + collector loop + $I_E = I_B + I_C$ (no β). Grounded emitter: I_B unchanged, $I_C = (V_{CC} - 0.2)/R_C$. With R_E the loops couple: solve 2×2
Edge of saturation	$I_C = \beta I_B$ **and** $V_{CE} = 0.2$ at once
β range	$I_C = \dfrac{\beta(V_{Th} - V_{BE})}{R_{Th} + (\beta+1)R_E}$: if $(\beta+1)R_E \gg R_{Th}$, $I_C \approx (V_{Th} - V_{BE})/R_E$ (stable); fixed bias $I_C \propto \beta$. To **avoid** saturation check β_{max}; to **guarantee** saturation use β_{min}. Quote both ends
pnp	$V_{EB} = 0.7$ ($V_B = V_E - 0.7$); I_B and I_C flow **out**, I_E in; active check $V_{EC} > 0.2$ V. Work in magnitudes; draw real arrows before KVL
Two BJTs, direct coupled	solve the stage with a self-contained base loop first; shared node ($V_{B2} = V_{C1}$ or $V_{E1} = V_{E2}$); npn stage 2: R_{C1} carries $I_{C1} + I_{B2}$; pnp stage 2: $I_{C1} - I_{B2}$ (lecture neglects I_{B2}: say so); check both states
Diode-connected BJT (B tied to C)	always active, drops one V_{BE}; its β is irrelevant
Divider design (Boylestad)	$V_E \approx 0.1V_{CC}$; $R_E = V_E/I_E$; $R_C = (V_{CC} - V_{CE} - V_E)/I_C$; $R_2 \leq \beta_{min}R_E/10$; $V_B = V_E + 0.7$; $R_1 = R_2(V_{CC} - V_B)/V_B$; standard values, then re-check with the exact method
Design from targets	$I_B = I_C/\beta$, $I_E = I_C/\alpha$, each $R = V_R/I_R$, starting with the loop that has one unknown

CHECK VALUES. WS5 Q10 (pnp, base grounded, 2.7 V via 1.5 k, collector 20 k to −20 V, β = 100): active guess gives $V_{EC} < 0$; saturated with $I_E = 1.333$, $I_C = 1.025$, $I_B = 0.308$ mA ($I_C < \beta I_B = 30.8$ mA). (The course's printed 0.925 mA is wrong.) WS5 Q11: cut-off ($I_E = -0.7$ mA in the active guess). WS5 Q12 ($I_B = 10$ µA, 9 k to 12 V): β = 150, $R_E = 1$ k saturated, $I_C = 1.179$, $I_E = 1.189$ mA; always saturated for $100 \leq \beta \leq 200$ if $R_E \geq 2.77$ kΩ. WS5 Q13: $I_{C,sat} = 4.8$ mA, $I_{B,min} = 24$ µA, $V_{BB} = 3.1$ V (β = 200; for β = 100: 48 µA, 5.5 V). WS5 Q16 (collector feedback, $80 \leq \beta \leq 120$): $V_o = 5.34$ V and 4.16 V. WS5 Q17: $R_C = 0.58$ kΩ (course 0.59). WS5 Q18 (α = 0.985, $V_{BE} = 0.3$): $R_1 = 53.4$ kΩ, $R_C = 2.97$ kΩ (46.8 kΩ if I_B is included in R_1). WS5 Q19: $R_C = 12.87$ kΩ, $R_B = 170$ kΩ. **PT Q8:** $V_{E1} = V_{E2} = 1.8$ V, $V_{B2} = 1.1$ V, $I_{B2} = 10$ µA, $I_{C2} = 0.5$ mA, $V_{C2} =$ **0.5 V** ($V_{EC2} = 1.3$ V; β_1 unused). Week 5 direct-coupled = WS6 Q16 (14 V, 33 k/10 k, 6.8 k, 2.2 k, β = 100; EF 1.2 k, β = 50): $I_{B1} = 11.1$ µA, $I_{C1} = 1.11$ mA, $V_{B2} = 6.44$, $V_{E2} = 5.74$ V, $I_{E2} = 4.78$ mA, $V_{CE2} = 8.26$ V (slide: 8.32 V, its 229.2 kΩ should be 229.9 kΩ; with I_{B2} included: 4.30 mA, 8.83 V).

BJT small-signal parameters and models

Parameter	Formula	Note
Transconductance	$g_m = I_C/V_T$	uses I_C
Emitter resistance	$r_e = V_T/I_E = 26\ \text{mV}/I_E = \alpha/g_m$	uses I_E; $g_m r_e = \alpha \approx 1$
Base input resistance	$r_\pi = \beta/g_m = V_T/I_B = (\beta+1)r_e \approx \beta r_e$	
Output resistance	$r_o = V_A/I_C = 1/h_{oe}\ (= 1/g_{os})$	drop only if $r_o \geq 10R_C$ (say so)
h-parameters (CE)	$h_{ie} = \beta r_e$, $h_{fe} = \beta_{ac}$, $h_{oe} = 1/r_o$, $h_{re} \approx 0$	$V_i = h_{11}I_i + h_{12}V_o$, $I_o = h_{21}I_i + h_{22}V_o$
Validity	$\lvert v_{be}\rvert \lesssim 10$ mV ($\ll V_T$): $i_C \approx I_C(1 + v_{be}/V_T)$	DC Q-point first
Gain ceiling	$\lvert A_v\rvert = g_m R_C = V_{R_C}/V_T$	6 V across R_C gives at most 231

Models (all the same circuit): hybrid-π: r_π B–E with v_π (+ at B), $g_m v_\pi$ C→E, r_o C–E. r_e model: βr_e B–E, βI_b C→E, r_o C–E. Since $v_\pi = i_b r_\pi$, $g_m v_\pi = g_m r_\pi i_b = \beta i_b$. T model: r_e in the emitter, αi_e (or $g_m v_\pi$) C→B; use it with an emitter resistor or a CB stage. npn and pnp have the same small-signal model.

AC EQUIVALENT: (1) DC sources to ground (V_{CC} becomes ground, so R_C and both R_1, R_2 go to ground: $R' = R_1 \parallel R_2$); (2) all coupling and bypass capacitors short; (3) delete anything shorted (a bypassed R_E vanishes); (4) redraw, insert the model. **Totals:** $v_C(t) = V_C + A_v v_{be}(t)$, $i_B(t) = I_B + v_{be}/r_\pi$.

Impedance reflection: a resistance R in the emitter looks like $(\beta+1)R$ from the base ($Z_b = \beta r_e + (\beta+1)R \approx \beta(r_e + R)$); a resistance R at the base looks like $R/(\beta+1)$ from the emitter.

CHECK VALUES. WS6 Q1 ($I_C = 1$ mA, β = 100, 25 mV): $g_m = 40$ mA/V, $r_e = 24.8\ \Omega$, $r_\pi = 2.5$ kΩ (the solution's "0.5 mA" is a typo). WS6 Q2 ($g_m = 20$ mA/V): $r_\pi = 5$ kΩ, $r_e = 49.5\ \Omega$. WS6 Q3 (1 mA, 10 kΩ, 15 V): $A_v = -400$, $v_C = 5 - 2\sin\omega t$ V, $i_B = 10 + 2\sin\omega t$ µA. WS6 Q4 (CB, T model, $r_e = 125\ \Omega$, 10 kΩ): $v_c/v_i = +80$. WS6 Q5: $v_y/v_{sig} = -76.5$.

Zi, Zo, Av and Ai for every BJT configuration

Table A: r_o ignored ($r_o \geq 10R_C$, or not given). $R' = R_1 \parallel R_2$. Unloaded.

Configuration	Z_i	Z_o	A_v	A_i (s
CE fixed bias	$R_B \parallel \beta r_e$	R_C	$-R_C/r_e$	$\approx \beta$
CE voltage divider, R_E bypassed	$R' \parallel \beta r_e$	R_C	$-R_C/r_e$	$\approx \frac{}{R}$
CE, R_E unbypassed (emitter bias or divider)	$R_B \parallel Z_b$ (or $R' \parallel Z_b$), $Z_b = \beta r_e + (\beta+1)R_E \approx \beta(r_e + R_E)$	R_C	$-\frac{\beta R_C}{Z_b} \approx -\frac{R_C}{r_e + R_E} \approx -\frac{R_C}{R_E}$	$\approx \frac{}{R}$
CE, partly bypassed (R_{E1} unbypassed, R_{E2} bypassed)	$Z_b \approx \beta(r_e + R_{E1})$	R_C	$-\frac{R_C}{r_e + R_{E1}}$	as ab
Emitter follower (CC)	$R_B \parallel Z_b$, $Z_b \approx \beta(r_e + R_E)$	$R_E \parallel r_e \approx r_e$; with source R_s: $R_E \parallel \left(r_e + \frac{R_s \parallel R_B}{\beta}\right)$	$\frac{R_E}{R_E + r_e} \approx 1$	$\approx -$
Common base	$R_E \parallel r_e \approx r_e$	R_C	$+\frac{\alpha R_C}{r_e} \approx \frac{R_C}{r_e}$	$-\alpha \approx$
Collector feedback (R_F C to B)	$\frac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_C + R_F}} = \beta r_e \parallel \frac{R_F}{1 - A_v}$ (Miller)	$R_C \parallel R_F$	$-\frac{R_F}{R_C + R_F}\frac{R_C}{r_e} = -\frac{R_C \parallel R_F}{r_e} \approx -\frac{R_C}{r_e}$	$\approx R$

BJT amplifier configurations of Table A

Table B: with r_o (re-derived symbolically from the r_e model; the unbypassed row is Boylestad's exact form).

Configuration	Z_i	Z_o	A_v	When r droppe
CE fixed bias	$R_B \parallel \beta r_e$	$R_C \parallel r_o$	$-\frac{R_C \parallel r_o}{r_e}$	$r_o \geq 1$
CE divider, bypassed	$R' \parallel \beta r_e$	$R_C \parallel r_o$	$-\frac{R_C \parallel r_o}{r_e}$	$r_o \geq 1$
CE unbypassed (partial: $R_E \to R_{E1}$)	$Z_b = \beta r_e + \frac{(\beta+1) + R_C/r_o}{1 + (R_C + R_E)/r_o} R_E$	$\approx R_C$ for any r_o (slide 45)	$\frac{-\frac{\beta R_C}{Z_b}\left(1 + \frac{r_e}{r_o}\right) + \frac{R_C}{r_o}}{1 + R_C/r_o}$	$r_o \geq 1$: Table
Emitter follower	$Z_b = \beta r_e + (\beta+1)(R_E \parallel r_o)$	$R_E \parallel r_o \parallel r_e$	$\frac{(\beta+1)(R_E \parallel r_o)}{Z_b} \approx \frac{R_E \parallel r_o}{R_E \parallel r_o + r_e}$	almost
Common base	$R_E \parallel r_e$	$R_C \parallel r_o$	$\frac{R_C}{r_e}\frac{r_o + r_e}{r_o + R_C} \approx \frac{R_C \parallel r_o}{r_e}$	$r_o \geq 1$
Collector feedback	$\beta r_e \parallel \frac{R_F}{1 - A_v}$ (with this A_v)	$R_C \parallel R_F \parallel r_o$	$-\left(\frac{1}{r_e} - \frac{1}{R_F}\right)(R_F \parallel R_C \parallel r_o) \approx -\frac{R_C \parallel R_F \parallel r_o}{r_e}$	$r_o \geq 1$

Loading (R_L, R_s; slides 46–52). Definitions: $A_{v_{NL}} = V_o/V_i$ unloaded, $A_{v_L} = V_o/V_i$ with R_L, $A_{v_s} = V_o/V_s$. Always $|A_{v_{NL}}| > |A_{v_L}| > |A_{v_s}|$.

Method	Rule	Exact for
A: load inside	CE, CB, collector feedback: $R_C \to R_C \parallel R_L$ (also $\parallel r_o$). Follower: $R_E \to R_E \parallel R_L$, **in Z_b too**: $Z_b = \beta(r_e + R_E \parallel R_L)$. Collector feedback: use A_{v_L} in the Miller Z_i	always
B: two-port	$A_{v_L} = A_{v_{NL}} \dfrac{R_L}{R_L + Z_o}$, $A_{v_s} = \dfrac{Z_i}{Z_i + R_s} A_{v_L}$, $A_{i_L} = -A_{v_L} \dfrac{Z_i}{R_L}$	CE fixed, divider, unbypassed (Z_i independent of R_L, Z_o of R_s). Not for the follower or collector feedback unless you use the loaded Z_i and source-dependent Z_o
Measured (Prac 3)	$Z_i = \dfrac{R_s V_i}{V_s - V_i}$, $Z_o = \dfrac{V_{o(NL)} - V_{o(L)}}{V_{o(L)}} R_L$	

Z_o never includes R_L; Z_i never includes R_s. Bypassing R_E raises the gain about $(r_e + R_E)/r_e$ times but lowers Z_i and makes the gain depend on r_e (bias, temperature); leaving R_E in gives $\approx -R_C/R_E$, free of β and r_e. Partial bypass is the dial between them. CB: huge A_v but $Z_i \approx r_e$, so R_s divides most of the signal away.

CHECK VALUES. Week 5 slide 34 (fixed, 12 V, 470 k, 3 k, β = 100, $r_o = 50$ k): $I_E = 2.428$ mA, $r_e = 10.71\ \Omega$, $Z_i = 1.07$ kΩ, $A_v = -280.2$; with r_o: $Z_o = 2.83$ kΩ, $A_v = -264.3$. Slide 50 (same, $R_L = 4.7$ k, $R_s = 0.3$ k, $r_o = \infty$): $A_{v_L} = -171.0$, $A_{v_s} = -133.5$. Slide 39 (20 V, 470 k, 2.2 k, 0.56 k, β = 120): $r_e = 5.99\ \Omega$; bypassed $Z_i = 717\ \Omega$, $A_v = -367$; unbypassed $Z_i = 59.3$ kΩ, $A_v = -3.89$ (with $r_o = 40$ k: $Z_b = 64.1$ kΩ, $A_v = -3.85$, $Z_o = 2.20$ kΩ); $R_{E1} = 120\ \Omega$: $Z_i = 14.6$ kΩ, $A_v = -17.5$. Slide 41 (22 V, 56 k/8.2 k, 6.8 k, 1.5 k bypassed, β = 90): $V_B = 2.81$ V, $I_E = 1.41$ mA, $r_e = 18.5\ \Omega$, $Z_i = 1.35$ kΩ, $A_v = -368$ (with $r_o = 50$ k: $Z_o = 5.99$ kΩ, $A_v = -324$). WS5 Q8 circuit: $r_e = 10.7\ \Omega$, bypassed $Z_i = 1.08$ kΩ, $A_v = -308$; unbypassed $Z_i = 6.42$ kΩ, $A_v = -3.27$; with $R_L = 4.7$ k, $R_s = 0.6$ k: $A_{v_L} = -181$, $A_{v_s} = -116$, $A_i = 41.6$. WS6 Q6 (exact divider, β = 180, $h_{oe} = 30$ µS): $r_e = 18.9\ \Omega$, $V_B = 3.72$ V, $Z_i = 3.17$ kΩ, $A_v = -217$ (course −217.7). WS6 Q7 (CB, 6 V, 6.8 k, 4.7 k): $I_E = 0.779$ mA, $r_e = 33.4\ \Omega$, $Z_i = 33.2\ \Omega$, $Z_o = 4.7$ kΩ, $A_v = 140.8$. WS6 Q8 (collector feedback, 12 V, 3.9 k, 220 k, β = 120): $r_e = 13.08\ \Omega$, $Z_i = 508\ \Omega$, $Z_o = 3.83$ kΩ, $A_v = -293$ by slide 44 (the WS6 solution's −298 drops the $R_F/(R_C + R_F)$ factor); with $r_o = 40$ k: $A_v = -267$, $Z_i = 539\ \Omega$, $Z_o = 3.50$ kΩ. Emitter follower (12 V, 220 k, 3.3 k, β = 100): $r_e = 12.6\ \Omega$, $Z_i = 132$ kΩ, $Z_o = 12.6\ \Omega$ (22.4 Ω with $R_s = 1$ k), $A_v = 0.996$; with $R_L = 1$ k: $Z_i = 57.6$ kΩ, $A_{v_L} = 0.984$, $A_{v_s} = 0.967$, $|A_i| = 56.7$.

Multistage amplifiers (Week 5, with the Week 8 FET cascade)

Item	Formulas	Rule
Cascade gain	$A_{vT} = A_{v1}A_{v2}\cdots$; $Z_i = Z_{i1}$; $Z_o = Z_{o,\text{last}}$	each stage loaded by the **next** stage's Z_i
Two-port chain	$A_{vs} = \dfrac{Z_{i1}}{Z_{i1}+R_s}\,A_{vNL1}\,\dfrac{Z_{i2}}{Z_{i2}+Z_{o1}}\,A_{vNL2}\,\dfrac{R_L}{R_L+Z_{o2}}$	exact for CE and CS types
Z_{i2} (the load on stage 1)	BJT CE bypassed: $R_1 \parallel R_2 \parallel \beta r_e$. Unbypassed: $R_1 \parallel R_2 \parallel \beta(r_e + R_E)$. FET: R_G or $R_1 \parallel R_2$	include the bias divider
FET cascade	stage 1: $-g_{m1}(R_{D1} \parallel R_{G2})$; last stage: $-g_m R_D$ (or $\parallel R_L$)	R_{G2} is MΩ, so loading is tiny
Darlington	$\beta_D = \beta_1\beta_2$; $V_{BE} = 2 \times 0.7 = 1.4$ V (1.6 V if a datasheet gives it); $I_B = \dfrac{V_{CC} - V_{BE}}{R_B + \beta_D R_E}$	follower: $Z_i \approx R_B \parallel \beta_D R_E$, $A_i \approx \dfrac{\beta_D R_B}{R_B + \beta_D R_E}$, $A_v \approx 1$
Feedback pair (pnp–npn)	$I_{B1} = \dfrac{V_{CC} - V_{EB1}}{R_B + \beta_1\beta_2 R_C}$	one V_{BE} drop
Cascode (CE into CB)	$V_{B1} = \dfrac{R_3}{R_1+R_2+R_3}V_{CC}$, $V_{B2} = \dfrac{R_2+R_3}{R_1+R_2+R_3}V_{CC}$, $I_E = \dfrac{V_{B1} - 0.7}{R_E}$ (same in both), $V_{C1} = V_{B2} - 0.7$	$A_{v1} \approx -1$, $A_{v2} \approx +R_C/r_e$, total $\approx -R_C/r_e$. The benefit is bandwidth (low Miller effect), not gain
Direct coupled	$V_{B2} = V_{C1} = V_{CC} - I_{C1}R_{C1}$	valid if $I_{B2} \ll I_{C1}$; otherwise use KCL at the node
Current mirror	$I_{REF} = \dfrac{V_{CC} - V_{BE}}{R}$ (split supplies: $\dfrac{V_{CC} + V_{EE} - V_{BE}}{R}$); $I_o = \dfrac{I_{REF}}{1 + 2/\beta} \approx I_{REF}$	design: $R = \dfrac{V_{CC} - V_{BE}}{I_o(1 + 2/\beta)}$. The output BJT must stay active
Constant-current source	$I \approx I_E = (V_B - V_{BE})/R_E$, with V_B fixed by a divider or Zener	independent of the load

Method: Procedure 16.

> **CHECK VALUES:** Week 5 BJT example (20 V, 15 k/4.7 k, 2.2 k, 1 k bypassed, β = 200): $Z_{i2} = 936\ \Omega$, $A_{v1} = -104$, $A_{v2} = -347$, $A_{vT} = 36.1 \times 10^3$, and 24.6×10^3 with $R_L = 4.7$ k (slides, r_e rounded to 6.34 Ω; unrounded $r_e = 6.39\ \Omega$ gives 35.6×10^3 and 24.2×10^3). Prac 4 Fig. 1 (stage 2 **unbypassed**): $Z_{i2} = 3.52$ kΩ, $A_{v1} = -212$, $A_{v2} = -2.19$, total 463, and 316 with $R_L = 4.7$ k; $Z_{i1} = 941\ \Omega$, $Z_{o2} = 2.2$ kΩ. WS6 Q15 cascode: $V_{B1} = 4.43$ V, $V_{B2} = 11.48$ V, $I_E = 3.73$ mA, $V_{C2} = 14.40$ V, $V_{C1} = 10.78$ V, $r_e = 6.97\ \Omega$, $A_v = -215$, $V_o = -2.15$ V for 10 mV (course −214, −2.14 V). WS6 Q12: $V_{BB} = 0.7 + 5(0.015) = 0.775$ V. WS6 Q13 ($I_{E2} = 120$ mA, $\alpha_1 = 0.99$, $\alpha_2 = 0.98$): $I_{C2} = 117.6$ mA, $I_{E1} = I_{B2} = 2.4$ mA, $I_{C1} = 2.376$ mA, $I_{B1} = 24$ µA, $I_C = 119.98$ mA. WS6 Q17 feedback pair (12 V, 1.8 M, 220 Ω, β = 80, 160): $I_{B1} = 2.45$ µA, $I_{C1} = I_{B2} = 196$ µA, $I_{C2} = 31.3$ mA, $V_{B1} = 4.41$ V, $V_{C2} = V_{E1} = 5.1$ V. WS6 Q19 mirror: $I_B = 9.9$ µA, $V_B = -0.990$ V, $V_E = -1.690$ V, $V_C = 2.574$ V; $R = (10 + 9.3)/1\ \text{mA} = 19.3$ kΩ. Mirror with β = 100: $I_o/I_{REF} = 0.980$. Week 8 JFET cascade: $I_{DQ} = 2.78$ mA, $A_{v1} = A_{v2} = -6.24$ with $g_m = 2.6$ mS, $A_v = 38.9$, $V_o = 389$ mV (g_m unrounded, 2.64 mS, gives 40.0).

Weeks 6–7 FETs: devices and DC bias

FET device equations

Common to all FETs: $I_G = 0$, so $I_S = I_D$ and there is no drop across R_G.

Device	Equation (saturation)	n-channel range	p-channel version
JFET	$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$, inverse $V_{GS} = V_P\left(1 - \sqrt{I_D/I_{DSS}}\right)$	$V_P \le V_{GS} \le 0, V_P < 0$. $I_D = 0$ beyond V_P	$V_P > 0, 0 \le V_{GS} \le V_P$, $V_{DS} < 0$
D-MOSFET	same Shockley equation	$V_{GS} \ge V_P$. V_{GS} may be **positive**, giving $I_D > I_{DSS}$	$V_P > 0$; $V_{GS} < 0$ enhances
E-MOSFET	$I_D = k\,(V_{GS} - V_{GS(\mathrm{Th})})^2$, with $k = \frac{I_{D(\mathrm{on})}}{(V_{GS(\mathrm{on})} - V_{GS(\mathrm{Th})})^2}$ in mA/V²	on for $V_{GS} > V_{GS(\mathrm{Th})} > 0$, otherwise $I_D = 0$. **No** I_{DSS}	$V_{GS(\mathrm{Th})} < 0$, on for $V_{GS} < V_{GS(\mathrm{Th})}$
E-MOSFET inverse	$V_{GS} = V_{GS(\mathrm{Th})} \pm \sqrt{I_D/k}$	take $+$	take $-$

Transfer-curve sketch (Shockley parabola, concave up):

V_{GS}	0	$0.3V_P$	$0.5V_P$	V_P	**D-MOS extra:** $V_{GS} = +1$ **V (n)**
I_D	I_{DSS}	$\approx I_{DSS}/2$ (exact 0.49)	$I_{DSS}/4$	0	$I_{DSS}(1 - 1/V_P)^2$

Saturation (pinch-off) check: $V_{DS} \ge V_{GS} - V_P$ (JFET, D-MOS) or $V_{DS} \ge V_{GS} - V_{GS(\mathrm{Th})}$ (E-MOS). If it fails, the device is ohmic and the square-law Q-point is invalid. Ohmic region: $r_d = r_o/(1 - V_{GS}/V_P)^2$.

p-channel rule: every polarity is mirrored. Solve the circuit with the n-channel formulas using magnitudes ($|V_P|$, $|V_{DD}|$, I_D as a positive number). Then put the signs back: V_{GS} and V_{DS} negative for a p-MOSFET, V_{GS} positive for a p-JFET. In CMOS, $V_{GS,p} = V_i - V_{DD}$.

FET DC bias

Bias lines (gate–source KVL; plot them on the transfer curve):

Configuration	Bias line	Intercepts / notes
Fixed bias (JFET, D-MOS)	$V_{GS} = -V_{GG}$	vertical line. Read the battery polarity
Self-bias	$V_{GS} = -I_D R_S$	through the origin
Voltage divider (any FET)	$V_{GS} = V_G - I_D R_S$, with $V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$	$(V_G, 0)$ and $(0, V_G/R_S)$
Common gate, R_S to $-V_{SS}$	$V_{GS} = V_{SS} - I_D R_S$	$V_S = -V_{SS} + I_D R_S$
D-MOS, gate and source both at 0 V	$V_{GS} = 0$	$I_D = I_{DSS}$, $g_m = g_{m0}$
E-MOS drain feedback (R_G from D to G)	$V_{GS} = V_{DS} = V_{DD} - I_D R_D$	always saturated
E-MOS divider	$V_{GS} = V_G - I_D R_S$	as divider

FET DC bias configurations

Output side: $V_D = V_{DD} - I_D R_D$, $V_S = I_D R_S$, $V_{DS} = V_D - V_S = V_{DD} - I_D(R_D + R_S)$. With $-V_{SS}$: $V_{DS} = V_{DD} + V_{SS} - I_D(R_D + R_S)$.

Ready-expanded quadratics for the bias line $V_{GS} = V_G - I_D R_S$ (self-bias: $V_G = 0$; CG: $V_G \to V_{SS}$; drain feedback: $V_G \to V_{DD}$, $R_S \to R_D$):

$$\text{Shockley, with } u = 1 - \frac{V_G}{V_P}: \qquad \frac{I_{DSS} R_S^2}{V_P^2} I_D^2 + \left(\frac{2 I_{DSS}\, u\, R_S}{V_P} - 1\right) I_D + I_{DSS}\, u^2 = 0$$

$$\text{E-MOS, with } w = V_G - V_{GS(\mathrm{Th})}: \qquad k R_S^2 I_D^2 - (2 k R_S w + 1) I_D + k\, w^2 = 0$$

Root-selection rule: find V_{GS} for **both** roots. Keep the root with $V_P \le V_{GS} \le 0$ (JFET), $V_{GS} \ge V_P$ (D-MOS) or $V_{GS} > V_{GS(\mathrm{Th})}$ (E-MOS). The other root lies on the mirror half of the

parabola. Write one line saying why you rejected it.

Method: Procedure 17.

> **CHECK VALUES** (all recomputed): Week 7 divider ($V_{DD} = 16$, 2.1 M/270 k, $R_S = 1.5$ k, $I_{DSS} = 8$, $V_P = -4$): roots 6.24 mA (reject, $V_{GS} = -7.53$) and **2.42 mA,** $V_{GS} = -1.80$ **V**. Week 7 CG ($R_S = 680\ \Omega$, 12 mA, −6 V): **3.84 mA, −2.61 V**. Week 7 drain feedback ($k = 0.24$, $V_{DD} = 12$, $R_D = 2$ k, $V_{GS(\mathrm{Th})} = 3$): **2.79 mA,** $V_{DS} = 6.41$ **V**.

Week 8 FET small-signal analysis

Transconductance g_m and r_d

Device / data	g_m	Note
JFET, D-MOS (maximum)	$g_{m0} = \dfrac{2I_{DSS}}{\lvert V_P \rvert}$	at $V_{GS} = 0$
JFET, D-MOS at Q	$g_m = g_{m0}\left(1 - \dfrac{V_{GSQ}}{V_P}\right) = g_{m0}\sqrt{\dfrac{I_{DQ}}{I_{DSS}}}$	D-MOS with $V_{GSQ} > 0$ gives $g_m > g_{m0}$
E-MOS	$g_m = 2k\,(V_{GSQ} - V_{GS(\mathrm{Th})}) = 2\sqrt{k\,I_{DQ}}$	no g_{m0}
Any square-law FET	$g_m = \dfrac{2I_{DQ}}{\lvert V_{GSQ} - V_{\mathrm{off}} \rvert}$, $V_{\mathrm{off}} = V_P$ or $V_{GS(\mathrm{Th})}$	quick cross-check
Graphical	$g_m = \Delta I_D / \Delta V_{GS}$, tangent at Q	the chord centred on Q is exact for a parabola
Datasheet	$g_m = g_{fs} = y_{fs}$, $r_d = 1/g_{os} = 1/y_{os}$	$r_d = \Delta V_{DS}/\Delta I_D$ at constant V_{GS}
Sedra (WS9 Q6–8)	$V_{ov} = V_{GS} - V_t$, $I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{ov}^2$, $g_m = \mu_n C_{ox}\frac{W}{L}V_{ov} = \dfrac{2I_D}{V_{ov}}$	course $k = \frac{1}{2}\mu_n C_{ox}(W/L)$. If W/L is asked, the given "k" is $\mu_n C_{ox}$

g_m against I_D: $I_{DSS}/2$ gives $0.707 g_{m0}$; $I_{DSS}/4$ gives $0.5 g_{m0}$; $g_m = 0$ at V_P. Device alone: $Z_o = r_d$, $A_{v(\mathrm{FET})} = -g_m r_d$ (WS8 Q9: 4.5 mS, 25 µS gives 40 kΩ and −180).

FET ac analysis: Z_i, Z_o, A_v (the FET ac table)

Model: gate–source open circuit; current source $g_m V_{gs}$ from drain to source, with r_d in parallel. The same model serves JFET, D-MOS and E-MOS. All entries below were re-derived symbolically from this model (Boylestad Table 8.2).

Configuration	Z_i	Z_o exact (with r_d)	Z_o, r_d ignored	A_v exact (with r_d)	A_v, r_d ig
CS fixed bias	R_G	$R_D \parallel r_d$	R_D	$-g_m(r_d \parallel R_D)$	$-g_m R_D$
CS self-bias, R_S bypassed	R_G	$R_D \parallel r_d$	R_D	$-g_m(r_d \parallel R_D)$	$-g_m R_D$
CS self-bias, R_S **unbypassed**	R_G	$R_D \dfrac{1 + g_m R_S + R_S/r_d}{1 + g_m R_S + (R_S + R_D)/r_d}$	R_D	$\dfrac{-g_m R_D}{1 + g_m R_S + (R_D + R_S)/r_d}$	$\dfrac{-g_m R}{1 + g_m}$
CS voltage divider (JFET or E-MOS), R_S bypassed	$R_1 \parallel R_2$	$R_D \parallel r_d$	R_D	$-g_m(r_d \parallel R_D)$	$-g_m R_D$
Source follower (CD)	R_G (or $R_1 \parallel R_2$)	$r_d \parallel R_S \parallel \dfrac{1}{g_m}$	$R_S \parallel \dfrac{1}{g_m}$	$\dfrac{g_m(r_d \parallel R_S)}{1 + g_m(r_d \parallel R_S)}$	$\dfrac{g_m R_S}{1 + g_m}$
Common gate	$R_S \parallel \dfrac{r_d + R_D}{1 + g_m r_d}$	$R_D \parallel r_d$	R_D	$\dfrac{g_m R_D + R_D/r_d}{1 + R_D/r_d}$	$+g_m R_D$
E-MOS drain feedback (R_F from D to G)	$\dfrac{R_F + r_d \parallel R_D}{1 + g_m(r_d \parallel R_D)}$	$R_F \parallel r_d \parallel R_D$	$R_F \parallel R_D \approx R_D$	$-\left(g_m - \frac{1}{R_F}\right)(R_F \parallel r_d \parallel R_D)$	$-g_m(R_F$

FET amplifier configurations of the FET ac table

Notes on the table:

- With r_d ignored, CG $Z_i \to R_S \parallel 1/g_m$ and drain-feedback $Z_i \to R_F/(1 + g_m R_D)$.
- D-MOS uses the same rows as the JFET. The slides use "$r_d \geq 10R_D$" as the condition for dropping r_d.
- **Phase:** CS and drain feedback invert (180°). The source follower and CG are in phase. Follower $A_v < 1$ always.
- **Loading:** put R_L in parallel with the output resistor (R_D, or R_S for the follower), for example $A_{vL} = -g_m(r_d \parallel R_D \parallel R_L)$. Then $A_{vs} = \frac{Z_i}{Z_i + R_{sig}} A_{vL}$. Z_o never includes R_L, and Z_i never includes R_{sig}. R_{sig} matters little for a gate input but a lot for CG ($Z_i \approx 1/g_m$).
- **BJT ↔ FET:** $1/g_m$ plays the role of r_e. Unbypassed CS $\approx -R_D/(1/g_m + R_S)$ matches $-R_C/(r_e + R_E)$.
- **Design for a gain:** $R_D \parallel r_d = |A_v|/g_m$, so $R_D = 1/(g_m/|A_v| - 1/r_d)$. Choose a standard value, then recheck V_{DS} for saturation. Week 8: $|A_v| = 10$, $g_m = 5$ mS, $r_d = 50$ k gives $R_D = 2.08$ kΩ, so use 2 kΩ.
- **Sedra shortcuts:** follower $R_o = 1/g_m$, $G_v = \frac{R_L}{R_L + 1/g_m}$. CG matched to the source: $1/g_m = R_{sig}$, so $G_v = \frac{1}{2} g_m R_D$ (WS9 Q6: $g_m = 10$ mS, $I_D = g_m V_{ov}/2 = 1$ mA, $G_v = 10$). $W/L = g_m/(\mu_n C_{ox} V_{ov})$.

Method: Procedure 18.

> **CHECK VALUE:** Week 8 fixed-bias example ($I_{DSS} = 10$, $V_P = -8$, $V_{GSQ} = -2$, $y_{os} = 40$ µS, $R_D = 2$ k) gives $g_{m0} = 2.5$ mS, $g_m = 1.875$ mS, $r_d = 25$ k, $Z_o = 1.85$ k, $A_v = -3.47$ (slides −3.48 from g_m rounded to 1.88), and -3.75 without r_d.

Weeks 9–10 Differential amplifiers, CMRR and op-amps

Quantity	Formula
Difference, common-mode inputs	$V_d = V_{i1} - V_{i2}$, $V_c = \frac{1}{2}(V_{i1} + V_{i2})$ (the average)
Output	$V_o = A_d V_d + A_c V_c = A_d V_d \left(1 + \frac{1}{\mathrm{CMRR}} \frac{V_c}{V_d}\right)$
CMRR	$\mathrm{CMRR} = A_d / A_c$; $\mathrm{CMRR(dB)} = 20 \log_{10}(A_d / A_c)$; ratio $= 10^{\mathrm{dB}/20}$ (100 dB is 10^5)
Measuring	$A_d = V_o / V_d$ with $V_c = 0$; $A_c = V_o / V_c$ with the inputs tied
BJT pair DC	$I_E(\text{tail}) = \frac{V_{EE} - 0.7}{R_E}$, each $I_C \approx I_E/2$, $V_C = V_{CC} - I_C R_C$, $r_e = 26\ \mathrm{mV}/(I_E/2)$
BJT pair gains	single-ended output: $A_v = \frac{R_C}{2r_e}$; double-ended: $A_d = \frac{R_C}{r_e}$; $A_c = \frac{\beta R_C}{r_i + 2(\beta+1)R_E} \approx \frac{R_C}{2R_E}$

Large R_E (or a current source in the tail) makes $A_c \to 0$, so CMRR $\to \infty$.

Ideal op-amp rules: (1) no input current; (2) with negative feedback, $V_+ = V_-$ (virtual short; virtual ground if $V_+ = 0$).

Circuit	Recognise	Output
Inverting	V_1 through R_1 to −, R_f from output to −, + grounded	$V_o = -\frac{R_f}{R_1} V_1$
Non-inverting	V_1 into +, R_1 from − to ground, R_f from output to −	$V_o = \left(1 + \frac{R_f}{R_1}\right) V_1$
Unity follower	output tied to −, input to +	$V_o = V_1$
Summing (inverting)	several R_k into −, one R_f	$V_o = -R_f \sum_k \frac{V_k}{R_k}$
Integrator (textbook)	R in, C in feedback	$v_o = -\frac{1}{RC} \int v_1 \, dt$
Multistage	outputs feed inputs	multiply the gains, keeping signs (ideal stages do not load each other)

Unfamiliar circuit: set $V_- = V_+$ and write KCL at the − node with zero pin current. Check $|V_o|$ against the rails (saturation at about $V_{CC} - 1$ to 2 V). **Check values:** WS10 Q1 gives $A_d = 120$, $A_c = 20$, CMRR = 6 = 15.6 dB. WS10 Q2 ($V_d = 60$ mV, $V_c = 170$ mV, $A_d = 6000$) gives 365.1 mV for CMRR = 200 and 360.01 mV for CMRR = 10^5. Method: Procedure 19.

Weeks 9–10 Power amplifiers

$\eta = \frac{P_o(\mathrm{ac})}{P_i(\mathrm{dc})} \times 100\%$. Transistor dissipation $P_Q = P_i - P_o$. Power in a resistor: $P = I_{rms}^2 R = \frac{I_p^2 R}{2} = \frac{I_{p\text{-}p}^2 R}{8}$ (same with V^2/R).

Class	DC input P_i	ac output P_o	Maximum	$\eta_{\max}$
A series-fed (R_C is the load)	$V_{CC}I_{CQ}$, fixed by the bias, not by the signal	$I_c^2(\text{rms})R_C$, with $I_c = \beta I_b$	peak I_c = **min** $(I_{CQ},\ V_{CC}/R_C - I_{CQ})$. Centred Q: $P_o = \dfrac{V_{CC}^2}{8R_C}$, $P_i = \dfrac{V_{CC}^2}{2R_C}$	25%
A transformer-coupled	$V_{CC}I_{CQ}$ ($V_{CEQ} \approx V_{CC}$, the swing reaches about $2V_{CC}$)	$\dfrac{(V_{CE\max} - V_{CE\min})(I_{C\max} - I_{C\min})}{8}$	$\eta = 50\left(\dfrac{V_{CE\max} - V_{CE\min}}{V_{CE\max} + V_{CE\min}}\right)^2\%$	50%
B push-pull	$\dfrac{2V_{CC}I_p}{\pi}$, with $I_p = V_L(p)/R_L$	$\dfrac{V_L^2(p)}{2R_L}$	$V_L(p) = V_{CC}$: $P_o = \dfrac{V_{CC}^2}{2R_L}$, $P_i = \dfrac{2V_{CC}^2}{\pi R_L}$	$\dfrac{\pi}{4}$ = 78.5%

Power amplifiers: class A series-fed, class A transformer-coupled, class B push-pull

Notes on the table:

- **Transformer:** $R'_L = a^2 R_L$, so $a = N_1/N_2 = \sqrt{R'_L/R_L}$. Also $V_2 = V_1/a$ and $I_2 = aI_1$. Combine the loads (series or parallel) first. WS10 Q9: four 16 Ω in parallel is 4 Ω, so $a = \sqrt{8000/4} = 44.7:1$.
- **Class B efficiency:** $\eta = \dfrac{\pi}{4}\dfrac{V_L(p)}{V_{CC}}$. Both transistors together: $P_{2Q} = P_i - P_o$, and each one dissipates $P_{2Q}/2$.
- **Class B worst-case dissipation** is not at full output. It is at $V_L(p) = 2V_{CC}/\pi = 0.636V_{CC}$, where $P_{2Q,\max} = \dfrac{2V_{CC}^2}{\pi^2 R_L}$.
- **Supply convention:** in these formulas V_{CC} is the supply for one half ($\pm V_{CC}$). A single-supply, capacitor-coupled push-pull uses $V_{CC}/2$.
- **Class AB** conducts between 180° and 360°, and its efficiency sits between those of A and B. **Class C** conducts for less than 180° and is for tuned (RF) circuits only.

Method: Procedure 20.

> **CHECK VALUES:** WS10 Q6 ($V_{CC} = 18$, $R_B = 1.2$ k, $R_C = 16\ \Omega$, $\beta = 40$, $I_b = 5$ mA rms): $I_{CQ} = 577$ mA, $P_i = 10.38$ W, $P_o = 0.64$ W. Q7 ($R_B = 1.5$ k): $P_i = 8.30$ W. Q8: peak $I_c = I_{CQ} = 461$ mA, $P_o(\max) = 1.70$ W (not 2.53 W). Q10 ($V_{CC} = 25$, 8 Ω): $P_i(\max) = 49.7$ W, $P_o(\max) = 39.1$ W, $\eta = 78.5\%$, $P_{2Q,\max} = 15.8$ W.

Procedures

Write the formula in symbols, substitute, then give the answer with units. State the model or state you assume, and finish with a one-line check. Those lines carry the method marks.

Procedure 1: Thévenin at a bias node (PT Q7 (a), WS5 Q14–15)

1. Remove the device (base, load) and leave node X open.
2. List every branch from X as (supply, resistor). Ground is a 0 V supply.
3. R_{Th}: short every supply; all branch resistors end up in parallel.
4. $V_{Th} = \sum(V_k/R_k)/\sum(1/R_k)$, or superposition one supply at a time.
5. Check that V_{Th} lies between the extreme supplies. Redraw V_{Th} in series with R_{Th} and reconnect the device.

Procedure 2: carrier concentration and resistance of a bar (WS2 Q2–Q7)

1. n_i: 1.5×10^{10} at 300 K, otherwise $BT^{3/2}e^{-E_g/2kT}$ with k in eV/K and T in K.
2. Donors give n-type, acceptors p-type. If doping $\geq 10n_i$: majority = doping, minority = n_i^2 /doping. Otherwise use the exact quadratic.
3. $\sigma = q(n\mu_n + p\mu_p)$, $\rho = 1/\sigma$.
4. Convert lengths to cm. A is the cross-section perpendicular to the current. $R = \rho L/A$.
5. If a voltage is given: $I = V/R$, $E = V/L$, $v_d = \mu E$.

Procedure 3: diode exponential numbers (PT Q2, WS2 Q8–Q10, WS3 Q1–Q4)

1. $T_K = 273 + T$, $V_T = kT_K/q$ to 4 s.f.
2. Pick the form. I_S, n, I_D given: $V_D = nV_T \ln(I_D/I_S)$. V_D given: full Shockley. Change in current: $\Delta V_D = nV_T \ln(I_2/I_1)$, showing that I_S cancels. Two points: divide for n, back-substitute for I_S, check with the other point.
3. Say "$I_D \gg I_S$, so the −1 is negligible".
4. Sanity check: $2.303\, nV_T$ per decade.
5. Temperature: $I_S \times 2^{\Delta T/10}$; V_D falls 2.5 mV/°C.

Procedure 4: diode circuit, assume and verify (PT Q3, WS3 Q15–Q17)

1. **Model.** Write the one named: 1st (ideal), 2nd (V_D), 3rd ($V_D + r_D$).
2. **Guess** each state and write it ("D1 ON, D2 OFF"). Anode nearer the more positive source: probably ON.
3. **Redraw and solve.** ON diodes become their model branch, OFF diodes open. Find i_d (anode → cathode) for each ON diode and $v_d = V_{anode} - V_{cathode}$ for each OFF diode. An OFF branch carries no current, so its resistors drop nothing but its batteries still count. Thévenin the rest of the circuit if there is one diode.
4. **Verify.** ON needs $i_d > 0$; OFF needs $v_d < V_D$ (ideal: < 0). If a check fails, flip that diode and repeat. Write the sentence: "$I = 15.3$ µA > 0, so all three conduct."
5. Only then compute V_o. For several ON diodes feeding one node, add KCL as the third equation.

Procedure 5: rectifier with a capacitor filter (WS3 Q9, Q14, Q20–21)

1. $V_m = \sqrt{2}V_{s(rms)}$ (centre tap: half-winding). $V_P = V_m - V_D$ or $V_m - 2V_D$ (bridge).
2. Write "half-wave" or "full-wave". $V_r = V_P/(fCR)$ or $V_P/(2fCR)$; rearrange for C when designing. If a ripple factor is given, convert: $V_{r(pp)} = 2\sqrt{3}\, rV_{dc}$.
3. $V_{O(dc)} = V_P - V_r/2$, $I_L = V_{O(dc)}/R$.
4. $\omega\Delta t = \sqrt{2V_r/V_P}$ rad; fraction $\omega\Delta t/2\pi$.
5. i_{Dav}, i_{Dmax} (formula-sheet half-wave forms; full-wave forms in the filter table).
6. PIV (half-wave with C: $2V_m$), output frequency, diode rating $\geq 1.5\times$ PIV.

Procedure 6: Zener regulator (PT Q4, WS3 Q10, Q13, Q22)

1. *Analysis:* assume breakdown. Ideal: $V_O = V_Z$, $I_S = (V_s - V_Z)/R_S$, $I_L = V_Z/R_L$, $I_Z = I_S - I_L$. With r_z: the single nodal equation for V_O.
2. Check $I_Z > 0$. If not, the Zener is off and $V_O = V_s R_L/(R_L + R_S)$.
3. $P_Z = V_O I_Z$.
4. *Design:* $V_Z = V_L$. Size R_S at lowest supply and heaviest load: $R_S = (V_{s,min} - V_Z)/(I_{L,max} + I_{Z,min})$.
5. Check the other extreme (highest supply, no load): $I_{Z,max}$ and $P_{Z,max} = V_Z I_{Z,max}$. That is often the last mark.
6. *Load range:* $R_{L,min}$ from $I_{R,min} - I_{Z,min}$; $R_{L,max}$ from $I_{R,max} - I_{Z,max}$ (∞ if ≤ 0).

Procedure 7: clipper or clamper (PT Q5, WS3 Q11–Q12, Q18)

1. Mark where v_o is taken and trace the only direction current can flow through each diode.
2. Assume OFF; write v_d in terms of v_i; set $v_d = V_D$ (or 0): v_{iT}.
3. One linear equation for v_o per region (KVL with battery signs as you meet them).
4. Square wave: evaluate both levels. Sine: find the crossing of v_{iT}.
5. Sketch v_o under v_i, same time axis, levels labelled; or draw the transfer characteristic (one line per state, meeting at v_{iT}).
6. Clamper: find the half-cycle that charges C (V_C = peak $-V_D$); in the other half $v_o = v_i \pm V_C$; check that peak-to-peak is unchanged.

Procedure 8: BJT DC bias with the state given (assume active) (PT Q6–Q7, WS5 Q3–Q9)

1. Thévenin the base (Procedure 1) if two resistors or two supplies meet there. For a divider, first write "$\beta R_E = \ldots$, $10R_2 = \ldots$, so the approximation is / is not valid".
2. Redraw with the active model (0.7 V B–E, βI_B C→E). A "redraw" part earns marks for every labelled element.
3. Base loop with $I_E = (\beta + 1)I_B$: $I_B = (V_{Th} - 0.7)/(R_{Th} + (\beta + 1)R_E)$. Approximate divider: $I_E = (V_B - 0.7)/R_E$.
4. $I_C = \beta I_B$, $I_E = (\beta + 1)I_B$.
5. Output loop: $V_{CE} = V_{CC} - I_C R_C - I_E R_E$; node voltages V_E, V_C, V_B.
6. Check: "$I_B > 0$ and $V_{CE} = \ldots > 0.2$ V, so active." If V_{CE} is negative, go to Procedure 9 step 5.

Procedure 9: BJT state unknown (guess, analyse, check) (WS5 Q9–Q15)

1. Simplify: Thévenin the base, and the collector if it has two supplies.

2. Guess: EBJ clearly forward biased → active; base undriven or reverse biased → cut-off.
3. Analyse with that model: active needs I_B, V_{CE}; saturation needs I_B, I_C; cut-off needs V_{BE}.
4. Check: active $I_B > 0$, $|V_{CE}| > 0.2$; saturation $I_B, I_C > 0$, $I_C < \beta I_B$; cut-off $|V_{BE}| < 0.7$.
5. Redirect: $V_{CE} \leq 0.2$ → saturation ($V_{CE} = 0.2$, solve base and collector loops with $I_E = I_B + I_C$); $I_B \leq 0$ → cut-off.
6. Quote the forced β (I_C/I_B) for saturation.

Procedure 10: β range or guaranteed saturation (WS5 Q12–Q13, Q16)

1. Write V_{CE} (or I_C) as a function of β with the active model.
2. "Stays active for all β": impose $V_{CE} \geq 0.2$ at β_{max}.
3. "Always saturated": at the edge, $I_C = \beta I_B$ and $V_{CE} = 0.2$, at β_{min}.
4. Solve the inequality for the unknown; state which β is the worst case and why. A negative resistance means "no value works": say so.
5. Range given: compute and quote both ends.

Procedure 11: load line and Q-point on printed characteristics (PT Q6)

1. $V_{CE} = V_{CC} - I_C(R_C + R_E)$; intercepts V_{CC} and $V_{CC}/(R_C + R_E)$; rule the line.
2. I_{BQ} from the base loop (with $(\beta + 1)R_E$).
3. Q where the line meets I_{BQ} (interpolate between printed curves).
4. Write the I_{CQ} and V_{CEQ} you read, then $\beta_{dc} = I_{CQ}/I_{BQ}$ from your readings. Quote the analytical values as a cross-check.

Procedure 12: bias design (WS5 Q4–Q6, Q17–Q19, Lecture 4)

1. List the targets; a stated $V_{CE} > 0.2$ V means active.
2. Currents first: $I_B = I_C/\beta$, $I_E = I_C/\alpha$.
3. Each resistor $R = V_R/I_R$, starting with the loop that has one unknown. Divider: rules in the state/design table, and say whether the base current was neglected.
4. Standard values, then re-analyse (exact method) and quote the % error.

Procedure 13: two-BJT DC circuits (PT Q8, WS6 Q12–Q17)

1. Draw real current arrows on every BJT (pnp: I_B, I_C out, I_E in).
2. Start with the transistor whose base loop is self-contained, or with a diode-connected BJT (fixes a level one V_{BE} away).
3. Carry the shared node into the next stage; include I_{B2} in the node KCL if accuracy matters, and say whether you did.
4. Use each transistor's own β; a β that only sets a level never appears in the answer.
5. Check every transistor's state.

Procedure 14: BJT ac analysis (r_e, Z_i, Z_o, A_v) (Week 5 examples, WS6 Q6–Q8)

1. DC Q-point first (capacitors open), with the right bias formula; get I_E; check active.
2. $r_e = 26\ \text{mV}/I_E$ to 3–4 s.f.; $r_\pi = \beta r_e$, $g_m = I_C/V_T$, $r_o = V_A/I_C$ or $1/h_{oe}$ if needed.
3. AC equivalent: V_{CC} to ground, capacitors shorted, shorted resistors deleted, model inserted.
4. Name the configuration: in at base → CE or follower; in at emitter → CB; out at collector → CE/CB; out at emitter → follower; R_F from collector to base → collector feedback. Mark what is in the emitter at AC: 0, R_E, R_{E1} or $R_E \parallel R_L$.
5. Read Z_i, Z_o, A_v from Table A; use Table B if r_o is given and $r_o < 10R_C$. Give both answers if the question supplies r_o.
6. Loading (Method A, or two-port where exact), then A_{v_s}, then $A_i = -A_{v_L} Z_i/R_L$ if asked.
7. State the phase. Sanity: $|A_{v_s}| < |A_{v_L}| < |A_{v_{NL}}|$; follower just under 1; unbypassed $\approx R_C/R_E$.

Procedure 15: derive a small-signal expression (WS6 Q9–Q10)

1. Draw the hybrid-π or r_e model with every resistor, labelled B, C, E.
2. $v_{out} = -\beta i_b \times$ (everything from collector to AC ground, including the next stage's r_π).
3. $v_{in} = i_b[R_B + r_\pi + (1 + \beta)R_{emitter}]$.
4. Divide; simplify only at the end; check that it collapses to the standard result when the extra resistors go to 0 or ∞.

Procedure 16: two-stage cascade (Week 5, Prac 4, Week 8)

1. DC bias each stage separately (capacitor-coupled stages are independent; direct-coupled stages are not). Get r_e or g_m for each stage.
2. Find Z_{i2}, including stage 2's bias resistors.
3. Stage 2 gain with its own load (R_L, or none).
4. Stage 1 gain with Z_{i2} as its load: $R_{C1} \to R_{C1} \parallel Z_{i2}$ (or $R_{D1} \parallel R_{G2}$).
5. $A_{vT} = A_{v1}A_{v2}$ with signs (CE–CE or CS–CS is positive). Carry 3–4 s.f. and do not round stage gains.

6. $Z_i = Z_{i1}$, $Z_o = Z_{o2}$; then $A_{vs} = \frac{Z_i}{Z_i + R_s} A_{vT}$.
7. $V_o = A_{vT} V_i$. Check it against the supply swing (clipping).

Procedure 17: FET DC bias by the quadratic (Week 7, WS8)

1. Identify the device from the data. I_{DSS} and V_P mean a JFET or D-MOS. $V_{GS(\mathrm{Th})}$ and $I_{D(\mathrm{on})}$ mean an E-MOS: find k first.
2. Find V_G: $-V_{GG}$, 0, or the divider value. There is no current in R_G.
3. Write the bias line from the table.
4. Substitute it into the device equation and use the expanded form above. Solve with the calculator.
5. Compute V_{GS} for each root and apply the root-selection rule.
6. Output loop: V_D, V_S, V_{DS} (and $V_{DG} = V_D - V_G$ if asked).
7. Saturation check, stated in one line.
8. Variants. **Measured** V_S: $I_D = V_S / R_S$, $V_{GS} = V_G - V_S$, $I_{DSS} = I_D / (1 - V_{GS}/V_P)^2$. **Design** (target V_D): $I_D = (V_{DD} - V_D)/R_D$, invert the device equation to get V_{GS}, then $R_S = (V_G - V_{GS})/I_D$. Recheck saturation.

Procedure 18: FET ac analysis (Week 8, WS9)

1. DC first (Procedure 17) to get V_{GSQ} and I_{DQ}, even when the question gives only the circuit.
2. g_m from the g_m table with the right formula for the device. $r_d = 1/g_{os}$, or ∞ if not given (say so).
3. Draw the ac equivalent: DC supplies to ground, all capacitors short, delete shorted resistors.
4. Name the configuration by the signal path: in at gate and out at drain is CS (is R_S bypassed? look for C_S); in at gate and out at source is a follower; in at source and out at drain is CG; R_F from drain to gate is drain feedback.
5. Read Z_i, Z_o, A_v from the FET ac table. Use the exact column if r_d is given and $r_d < 10 R_D$.
6. Load: R_L in parallel with the output resistor, then the R_{sig} divider.
7. $V_o = A_v V_i$. State the sign or phase. Check $|V_o|$ against the available swing.

Procedure 19: differential amplifier or op-amp circuit (WS10 Q1–Q5)

1. Op-amp: check for negative feedback, then $V_- = V_+$ and zero pin current.
2. Match a standard circuit from the op-amp table. If none fits, write KCL at the – node.
3. Multistage op-amps: multiply the gains with their signs.
4. Check $|V_o|$ against the rails.
5. Differential inputs: $V_d = V_{i1} - V_{i2}$, $V_c = \frac{1}{2}(V_{i1} + V_{i2})$, CMRR as a ratio ($10^{\mathrm{dB}/20}$), $V_o = A_d V_d (1 + V_c / (\mathrm{CMRR}\, V_d))$.

Procedure 20: power-amplifier efficiency (WS10 Q6–Q10)

1. Identify the class: load in the collector (series A), transformer primary in the collector (transformer A), or a complementary or push-pull pair (B).
2. Class A: DC bias first, $I_B = \frac{V_{CC} - 0.7}{R_B}$, $I_{CQ} = \beta I_B$, $V_{CEQ} = V_{CC} - I_{CQ} R_C$. Check that it is active.
3. P_i: $V_{CC} I_{CQ}$ (class A) or $2 V_{CC} I_p / \pi$ (class B).
4. P_o: convert the signal to rms, peak or p-p and use the matching form. For “maximum”, use the clipping limit (A) or $V_L(p) = V_{CC}$ (B).
5. $\eta = P_o / P_i$. It must not exceed 25%, 50% or 78.5%.
6. Dissipation: $P_Q = P_i - P_o$. For class B, give P_{2Q} and the per-transistor value, and the worst case if asked.

Which method when

The question shows or asks	Use
A node fed by two supplies or a divider, “Thévenin seen from the base”	Procedure 1 (superposition for V_{Th})
N_D, N_A, a bar’s dimensions, a temperature	Procedure 2 (n_i formula if $T \neq 300$ K)
“Exponential model”, n, I_S, currents decades apart	Procedure 3: $\Delta V_D = nV_T \ln(I_2/I_1)$
Diodes in a resistor network, “verify whether conducting”, a named PWL model	Procedure 4
“dc”, “static” resistance / “ac”, “dynamic”, “small-signal”	V_D/I_D / slope or nV_T/I_D
Two given points on a curve, “piecewise-linear equivalent”	$r_{av} = \Delta V/\Delta I$, 3rd model
$V_{s(rms)}$, PIV, ripple, “choose C”, diode current	Procedure 5
Zener, “maintain V_L”, load range	Procedure 6
“Determine v_o for the input shown”, battery in series with a diode	Procedure 7
Capacitor in series and diode to ground; “doubler”	clamper / doubler rules
Two of I_B, I_C, I_E, α, β	KCL, then the ratios
One base resistor; or one base resistor and R_E	fixed or emitter-stabilised bias (Procedure 8)
Divider at the base	write the $\beta R_E \geq 10R_2$ test, then exact or approximate
“Assume active”	Procedure 8 with the one-line check
“Find the mode/state”, nothing assumed	Procedure 9
A range of β, “always saturated”, “for what R”	Procedure 10
Printed I_C–V_{CE} curves, “draw the load line”	Procedure 11 (read the graph; analytical value as check)
“Select/choose R so that…”	Procedure 12
Arrow into the emitter, V_{EB}, negative rails	pnp conventions (magnitudes, real arrows)
Two BJTs, no coupling capacitor; base tied to collector	Procedure 13
“Find g_m, r_e, r_π”	parameter table (I_C for g_m, I_E for r_e)
“Find Z_i, Z_o, A_v” for a BJT	Procedure 14, Table A (Table B if $r_o < 10R_C$)
C_E across only part of R_E	DC uses $R_{E1} + R_{E2}$, AC uses R_{E1}
Output at the emitter with R_s and/or R_L	follower: $R_E \parallel R_L$ inside Z_b; Z_o with R_s/β
Signal into the emitter, base grounded	common base: $Z_i \approx r_e$, in phase
Resistor from collector to base, no capacitor	collector feedback: Miller Z_i
“Find an expression”, hybrid-π	Procedure 15
I_{DSS}, V_P and R_S or a divider; “find I_{DQ}”	Procedure 17 (the quadratic). Graphical only if it says “sketch” or “graphically”
Fixed bias, or D-MOS with gate and source at 0 V	no quadratic: plug V_{GS} straight into Shockley
A measured V_S, V_D or V_{DS}	KVL for I_D first, then V_{GS}, then the unknown device parameter
“Choose R_S / R_D to give…”	inverse device equation, then the bias line (design)
E-MOSFET data ($V_{GS(\mathrm{Th})}$, $I_{D(\mathrm{on})}$)	find k first; then $g_m = 2k(V_{GSQ} - V_{GS(\mathrm{Th})})$
A transfer curve drawn	graphical g_m (tangent); compare with the formula
V_{ov}, $\mu_n C_{ox}$, W/L (WS9 Q6–8)	Sedra rows of the g_m table and the FET ac notes

The question shows or asks	Use
Signal in at the gate	CS (out at drain) or follower (out at source)
Signal in at the source, gate grounded	common gate: small Z_i, so R_{sig} matters
r_d given and comparable to R_D	exact column of the FET ac table
R_{sig} and/or R_L given	loaded gain, then the input divider (BJT loading table, FET ac notes, or the two-port chain)
Two transistors, capacitor-coupled	Procedure 16
Darlington, cascode, feedback pair, current mirror	multistage table
Two BJTs with one shared emitter resistor	differential pair
Two inputs with A_d and CMRR	$V_o = A_d V_d(1 + V_c/(\mathrm{CMRR}\, V_d))$
Op-amp figure	identify the input pin, then the op-amp table (Procedure 19)
Power, efficiency or dissipation	Procedure 20
"Speakers appear as..."	$a = \sqrt{R_L'/R_L}$

Traps

Weeks 1–2 (circuit tools, semiconductors, diodes)

Treating a negative rail as ground in a Thévenin (PT Q7: 6 V instead of 2 V). Every supply is a branch.

Divider used with a load on the middle node. It is only valid for series elements.

Units in $R = \rho L/A$: µm and mm not converted to cm. **k in J/K inside n_i** (needs eV/K), or T in °C.

$n + p = n_i$ instead of $np = n_i^2$. Doped material described as charged (it is neutral).

V_T not recomputed for the stated temperature (20 °C is 25.27 mV).

n dropped from $\Delta V = nV_T \ln(I_2/I_1)$, or $\log_{10}$ used (PT Q2: 119 mV or 77.6 mV instead of 179 mV).

$R_D = V_D/I_D$ given when r_d (slope, nV_T/I_D) was asked, or the reverse.

No verification line after assuming diode states; OFF check written as i_d instead of v_d; r_D terms left out of the 3rd model; the shunt resistor at the input node ignored (PT Q3 pitfall: 0.333 mA).

Full-wave ripple without the 2 ($V_P/(2fCR)$ is not on the formula sheet). Bridge drops $2V_D$. Half-wave PIV doubles with a filter capacitor.

V_P quoted as the dc output when a ripple is given ($V_P - V_r/2$). Ripple factor 10 % is $V_{r(pp)} = 0.346V_{dc}$, not $0.1V_{dc}$.

Zener R_S sized at no load, or the no-load power check left out. A filtered supply uses the ripple valley as $V_{s,min}$.

Clipper diode direction or battery sign misread (PT Q5: 16 V or −9 V instead of 0 / −1 V). Trace the only direction current can flow first.

Weeks 3–5 (BJTs)

WS4 handout mode table has saturation and cut-off swapped. Use the lecture table.

α = β/(1+α) or $I_E = \beta I_B$. Use α = β/(β+1), $I_E = (\beta + 1) I_B$.

$(\beta + 1) R_E$ **forgotten** in the base loop (PT Q6: 71.5 µA instead of 35.3 µA). Load line drawn with R_C only.

Approximate divider used without writing the $\beta R_E \geq 10 R_2$ **test**, or used when it fails (WS5 Q7: 16 % error). Never valid with $R_E = 0$.

Negative V_{CE} **treated as a slip.** It means saturation: redo with $V_{CE} = 0.2$ V and do not use $I_C = \beta I_B$.

Wrong β for a range: avoid saturation at β_{max}, guarantee it at β_{min}.

pnp with npn polarity (PT Q8: $V_{B2} = 2.5$ V). $V_B = V_E - 0.7$; base and collector currents flow out.

V_{CE} **and** V_C **confused.** Equal only with a grounded emitter.

β reading: "β at the Q-point" is I_{CQ}/I_{BQ} from your graph readings, not the stated β quoted back.

AC built on a wrong Q-point. r_e uses I_E, g_m uses I_C. State V_T (26 or 25 mV).

Bypassed and unbypassed mixed up. Partial bypass: DC uses the whole R_E, AC only R_{E1}. Look for C_E.

r_o **silently dropped** when given and $r_o < 10 R_C$ (slide 41: −368 becomes −324). Unbypassed Z_o stays R_C even with r_o.

Follower Z_i **without** R_L, or $Z_o = R_E \parallel r_e$ when driven from R_s (add R_s'/β). Two-port loading used blindly on a follower or collector feedback.

CB given a minus sign, or R_s ignored in front of a 30 Ω input.

Collector feedback Z_i **taken as** R_F**.** It is $\beta r_e \parallel R_F/(1 - A_v)$ and depends on the load.

Loaded gain larger than unloaded. That is a sign or reciprocal error.

Cascade: stage 1 not loaded by Z_{i2} (with stage 2's bias resistors); stage gains rounded before multiplying; V_o not checked against the swing (Prac 4: 30 mV input would clip).

Darlington with one V_{BE}**.** Current mirror with $1/\beta$ in place of $2/\beta$.

Weeks 6–8 (FETs)

Sign inside Shockley: $1 - \frac{-2}{-4.5} = 0.556$, not 1.444. Only g_{m0} uses $|V_P|$. A JFET with $I_D > I_{DSS}$ means a sign error. Only a D-MOS with V_{GS} of the enhancing sign can exceed I_{DSS}.

Wrong or both roots kept. Test V_{GS} for each root and state which one you rejected and why.

I_{DSS} **or the JFET** g_m **used for an E-MOSFET.** Use k and $2k(V_{GSQ} - V_{GS(\mathrm{Th})})$.

A voltage across R_G**.** $I_G = 0$, so there is none. **The** V_{GG} **battery polarity** sets the sign of V_G (WS8 Fig. 1–2).

$V_S = I_D R_S$ **when** R_S **returns to** $-V_{SS}$**.** It is $-V_{SS} + I_D R_S$.

Skipping the saturation check. WS8 Q5 and WS9 Q4 give answers that fail it. Week 7 Example 2 sits on the edge.

Unbypassed R_S treated as bypassed, or the reverse. Look for C_S. The slide 53 entry $Z_o = R_D/(1 + g_m R_S)$ is wrong. Use $Z_o \approx R_D$ (exact form in the FET ac table).

Source follower $Z_o = R_S$. It is $R_S \parallel 1/g_m$, dominated by $1/g_m$. **Drain feedback $Z_i = R_F$.** It is $R_F/(1 + g_m R_D)$.

Sign of the gain. CS is negative (slide 40 drops it: it should be −8.05). CG and the follower are positive. CE–CE is positive.

R_L put in Z_o, or R_{sig} put in Z_i.

Unloaded stage gains multiplied in a cascade, or Z_{i2} without the bias resistors. Do not round stage gains before multiplying.

Unit slips: µS against mS, k in mA/V² used with I_D in A, MΩ in Z_i.

Weeks 9–10 (differential amplifiers, op-amps, power amplifiers)

V_c taken as the sum, or CMRR in dB put into the formula. Use the average and the ratio.

Non-inverting gain written as R_f/R_1. It is $1 + R_f/R_1$.

rms, peak and p-p mixed in power formulas. Class A P_i changes with R_B, not with the signal.

Centred-Q maximum ($V_{CC}^2/8R_C$) used when the Q-point is not centred. Take the smaller clipping limit.

Class B P_{2Q} quoted per transistor, or the maximum dissipation taken at full output.

Turns ratio inverted, or parallel speakers added in series.

Circuit tools and semiconductor physics

Cheat sheet · Day 01 · Sat 24 Oct · pack E1

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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E1 - CIRCUIT TOOLS + SEMICONDUCTOR PHYSICS
Divider (series only): Vi = Vin*Ri/Rsum   Current div (2 R): I1 = I*R2/(R1+R2)
Thevenin: Voc with load removed; RTh = R seen with V->short, I->open
  dep. sources present: RTh = Voc/Isc ; only dep.: RTh = Vtest/Itest
  IN = VTh/RTh, RN = RTh
Bias node X fed by (Vk,Rk), ground = 0 V branch:
  VTh = sum(Vk/Rk)/sum(1/Rk)    RTh = R1||R2||...
  Divider: VTh = VCC*R2/(R1+R2), RTh = R1||R2  -> then base KVL:
  VTh = IB*RTh + VBE + IE*RE
Nodal: ref node, KCL (leaving = source in), V-src between nodes -> supernode
Superposition: one indep. source at a time; dep. sources stay ON
Ideal op-amp: v+ = v-, i+ = i- = 0, no KCL at output; Vo limited by rails
---------------------------------------------------------------
ni = B T^1.5 exp(-Eg/2kT)  B=7.3e15, Eg=1.12 eV, k=8.62e-5 eV/K, T in K
ni(300K) = 1.5e10 cm^-3 ; ni^2 = 2.25e20
n0*p0 = ni^2 (equilibrium)
n-type: n ~ ND, p = ni^2/ND ; p-type: p ~ NA, n = ni^2/NA
Exact (n-type): n = ND/2 + sqrt((ND/2)^2 + ni^2), p = ni^2/n  (if ni >~ 0.1 ND)
sigma = q(n mu_n + p mu_p) (ohm-cm)^-1 ; rho = 1/sigma ; R = rho L/A (cm!)
vd = mu E, E = V/L ; J = sigma E ; D/mu = VT (Einstein)
mu: WS uses 1350/500, lecture slide 1300/500 -> use what is given
VT = kT/q, k = 1.38e-23 J/K, q = 1.6e-19 ; 27C -> 25.875 mV
1 um = 1e-4 cm ; 1 um^2 = 1e-8 cm^2 ; 1 mm = 0.1 cm
pn junction: open ckt ID (diffusion, majority) = IS (drift, minority)
  xn/xp = NA/ND (extends into lightly doped side), W ~ 0.1-1 um
  V0 = VT ln(NA ND / ni^2) ; FB: W shrinks, I exp ; RB: W grows, only IS
  IS doubles per 10 C ; doped material is electrically NEUTRAL
Group V donor -> n-type ; Group III acceptor -> p-type
```

Diode models

Cheat sheet · Day 03 · Mon 26 Oct · pack E2

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
DIODE MODELS (T2, Wk 1-2)
V_T = kT_K/q  (k=1.38e-23, q=1.6e-19; T_K=273+C)  27C->25.875mV, 25C->25.72, 20C->25
I_D = I_S(e^{V_D/nV_T} - 1) ~ I_S e^{V_D/nV_T} (forward);  reverse: I_D ~ -I_S
V_D = nV_T ln(I_D/I_S)       dV_D = nV_T ln(I2/I1)   [I_S cancels]
1 decade = 2.303 nV_T  (n=1: ~60 mV; n=1.5: ~89 mV)
Two points: n = (V2-V1)/(V_T ln(I2/I1)); I_S = I1 e^{-V1/nV_T}; check with point 2
Parallel identical diodes, same I: V falls by nV_T ln2 (~18 mV)
TEMP: V_D falls 2.5 mV/C (fixed I); I_S doubles per 10C: x 2^{dT/10}; V_T ~ T
R_D = V_D/I_D (dc, static)   r_d = dV/dI = nV_T/I_D (ac/small-signal, v_d < ~5 mV)
r_av = dV/dI between two points -> 3rd model (V_K + r_av). Knee V_K: Ge 0.3, Si 0.7,
MODELS  ON:  1 ideal = short | 2 = V_D source | 3 = V_D + r_D   (+ on anode side)
        OFF: open circuit in ALL models. 3rd model: v_d = V_D + i_d r_D
ASSUME-VERIFY: 1 model  2 guess each diode (write it)  3 redraw+solve: i_d for ON,
  v_d = V_anode - V_cathode for OFF (OFF branch: no I, resistors drop 0, batteries co
  4 check ON: i_d > 0 ; OFF: v_d < V_D (ideal: < 0). Fail -> flip that diode, resolve
Tip: Thevenin everything except the diode -> i_d = (V_Th - V_D)/(R_Th + r_D)
PT Q2: 0.179 V.  PT Q3: I = 15.3 uA, Vo = 3.15 V (0.7 V model would make them OFF!)
```

EDC · T3

Rectifiers, Zener regulators, clippers

Cheat sheet · Day 05 · Wed 28 Oct · pack E3

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
RECTIFIERS   Vm = √2·Vs(rms).  Vp = Vm − VD (HW, CT)  |  Vm − 2VD (bridge)
  Vdc(no C): HW Vm/π   FW 2Vm/π    f_out: HW f, FW 2f
  PIV: HW Vs (≈2Vm WITH C) | CT 2Vs − VD | bridge Vs − VD ; rate ≥1.5×PIV
  With rf: Im = Vm/(rf+R) [bridge 2rf+R]; HW Idc=Im/π, Irms=Im/2; FW 2Im/π, Im/√2
FILTER (CR >> T)   Vr(pp) = Vp/(fCR) = IL/(fC)   FULL-WAVE: use 2f
  Design: C = Vp/(f R Vr) [or 2f].  Vo(dc) = Vp − Vr/2,  IL = Vo/R
  ωΔt = √(2Vr/Vp) rad ; fraction = ωΔt/2π ; Δt = ωΔt/(2πf)
  HW: iDav = IL(1+π√(2Vp/Vr))   iDmax = IL(1+2π√(2Vp/Vr))
  FW: iDav = IL(1+π√(Vp/2Vr))   iDmax = IL(1+2π√(Vp/2Vr))
  Ripple factor r = Vr(rms)/Vdc, Vr(rms) = Vr(pp)/(2√3) -> 10% => Vr(pp)=0.346 Vdc
  %VR = (VNL − VFL)/VFL × 100
ZENER   VZ = VZ0 + rz·IZ ;  IS = (Vs−VO)/RS = IZ + IL
  Analyse: (VO−VZ0)/rz + VO/RL = (Vs−VO)/RS ; check IZ>0 ; PZ = VO·IZ
  Design: VZ = VL ; RS = (Vs,min − VZ)/(IL,max + IZ,min)   [PT Q4: 20 Ω]
          PZ,max = VZ[(Vs,max − VZ)/RS − IL,min]  (no load!)  [PT Q4: 2.4 W]
  Load range: RL,min = VZ/(IR,min − IZ,min) ; RL,max = VZ/(IR,max − IZ,max) (∞ if ≤0)
  Line reg = (rz||RL)/(RS + rz||RL) ; Load reg = −(rz||RS)
  Filtered supply: use the ripple VALLEY Vp − Vr as Vs,min
CLIPPER: assume OFF -> vd(vi) -> set vd = VD -> viT -> vo per region -> sketch under
  [PT Q5: +20/−5 V square, 4 V battery opposing: vo = 0 / −1 V]
CLAMPER: C charges to Vm, waveform shifts by Vm, same p-p. DOUBLER: ≈2Vm, PIV 2Vm
LED λ(nm) ≈ 1240/Eg(eV); Schottky low VF, fast; varactor Cj↓ as VR↑; photodiode reve
```

BJT operation and DC bias I

Cheat sheet · Day 06 · Thu 29 Oct · pack E4

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
BJT MODES (lecture table; the WS4 handout has sat/cut-off rows swapped)
  Active: EBJ fwd, CBJ rev | VBE=0.7, IC=bIB      | check IB>0, VCE>0.2
  Sat:    EBJ fwd, CBJ fwd | VBE=0.7, VCE=0.2     | check IB,IC>0, IC<bIB
  Cutoff: both rev         | IB=IC=IE=0           | check VBE<0.7
CURRENTS  IE=IC+IB   IC=bIB   IE=(b+1)IB   IC=aIE
  a=b/(b+1)   b=a/(1-a)   ICEO=ICBO/(1-a)=(b+1)ICBO
  b_dc = ICQ/IBQ at the Q-point   (b_ac = dIC/dIB, curve spacing)
BIAS (assume active, then CHECK)
  Fixed:     IB=(VCC-0.7)/RB                VCE=VCC-IC*RC
  Emitter:   IB=(VCC-0.7)/(RB+(b+1)RE)      VCE=VCC-IC*RC-IE*RE
  Divider exact: RTh=R1||R2, ETh=VCC*R2/(R1+R2)
                 IB=(ETh-0.7)/(RTh+(b+1)RE)
  Divider approx ONLY IF b*RE >= 10*R2  (write the test!)
                 VB=ETh, VE=VB-0.7, IC~IE=VE/RE, VCE~VCC-IC(RC+RE)
  Two supplies on the base -> superposition for VTh (PT Q7)
  Node voltages: VE=IE*RE, VC=VCC-IC*RC, VB=VE+0.7, VBC=VB-VC (<0 active)
LOAD LINE  VCE=VCC-IC(RC+RE):  (VCC,0) and (0, VCC/(RC+RE)=ICsat)
  Q = load line meets IB=IBQ curve; write the IC, VCE you read; b=ICQ/IBQ
  Design: VCC from VCE intercept, RC(+RE)=VCC/ICsat,
          RB=(VCC-0.7-IE*RE)/IBQ -> nearest standard value, quote % error
Negative VCE from an active calc => saturated, redo (pack E5)
```

BJT DC bias II

Cheat sheet · Day 08 · Sat 31 Oct · pack E5

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
BJT DC BIAS II (E5)
STATE LOOP: guess -> analyse -> check -> redo
 active: |VBE|=0.7, IC=bIB, IE=(b+1)IB    check IB>0, |VCE|>0.2
 sat:    |VBE|=0.7, |VCE|=0.2, IE=IB+IC   check IB,IC>0, IC<bIB (quote IC/IB)
 cutoff: IB=IC=IE=0                        check |VBE|<0.7
 VCE<=0.2 (negative!) -> sat ;  IB<=0 -> cutoff
 grounded emitter: IB same in both attempts, IC,sat=(VCC-0.2)/RC
 RE present: base & collector loops coupled in sat -> solve 2x2

GENERAL:  IB = (VTh - VBE) / (RTh + (b+1)RE)
 beta range: AVOID sat -> use b_max ; GUARANTEE sat -> use b_min (edge: IC=bIB, VCE=0
 negative R from inequality -> "no value works", explain

COLLECTOR FEEDBACK: IB=(VCC-VBE)/(RF+(b+1)(RC+RE)); RC carries IE
   VCE=VCC-IE(RC+RE)   (slides use b(RC+RE): IC~IE)
EMITTER FOLLOWER: IB=(VEE-VBE)/(RB+(b+1)RE); VCE=VEE-IE*RE
COMMON BASE: IE=(VEE-VBE)/RE; VCB=VCC-IC*RC; VCE=VEE+VCC-IE(RC+RE)

PNP: VB=VE-0.7; IB, IC OUT; IE IN; check VEC>0.2. Draw arrows first.
TWO STAGES: solve stage with own base loop first; shared node;
 npn 2nd stage: RC1 carries IC1+IB2 ; pnp 2nd stage: RC1 carries IC1-IB2
 diode-connected (B=C): always active, drops 0.7 V, its beta irrelevant
APPROX DIVIDER valid only if b*RE >= 10 R2 (never with RE=0)

DESIGN (VDB): VE=0.1VCC; RE=VE/IC; RC=(VCC-VCE-VE)/IC;
 R2 <= b_min RE/10; VB=VE+0.7; R1=R2(VCC-VB)/VB; standard values -> re-check exact
EARLY: iC=IS e^(vBE/VT)(1+vCE/VA); ro=(VA+VCE)/IC ~ VA/IC
```

BJT small-signal I

Cheat sheet · Day 10 · Mon 2 Nov · pack E6

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
BJT SMALL-SIGNAL I  (VT = 26 mV Boylestad / 25 mV Sedra - state which)
gm = IC/VT     r_pi = beta/gm = VT/IB = (beta+1) re ~ beta re
re = VT/IE = alpha/gm     ro = VA/IC = 1/hoe     gm*re = alpha ~ 1
Models: hybrid-pi (r_pi, gm v_pi, ro) == re model (beta re, beta Ib, ro)
        since gm v_pi = gm r_pi ib = beta ib.   T model: re in emitter, alpha ie C->E
h-params (CE): hie = beta re, hfe = beta, hoe = 1/ro, hre ~ 0
Small-signal valid: |vbe| < ~10 mV

AC EQUIVALENT: DC sources -> ground, all C -> short, delete shorted R, redraw, insert

ALWAYS: 1) DC Q-point (caps open) -> IE   2) re = 26mV/IE   3) AC circuit   4) Zi, Zo
CE FIXED BIAS:   IB=(VCC-0.7)/RB, IE=(b+1)IB
   Zi = RB || beta re     Zo = RC || ro     Av = -(RC||ro)/re  (~ -RC/re if ro >= 10RC)
CE VDB, RE BYPASSED:  test beta RE >= 10 R2 -> VB = R2 VCC/(R1+R2), IE=(VB-0.7)/RE
                      else Thevenin: IB = (ETh-0.7)/(RTh+(b+1)RE)
   Zi = R1||R2||beta re   Zo = RC || ro     Av = -(RC||ro)/re
Phase: minus sign = 180 deg.   Check: |Av| = gm RC = V_RC/VT
Totals: vC(t) = VC + Av vbe(t);  iB(t) = IB + vbe/r_pi
```

BJT small-signal II

Cheat sheet · Day 12 · Wed 4 Nov · pack E7

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
SMALL-SIGNAL BJT II  (always: DC Q-point -> IE -> re = 26 mV/IE)
Reflection: R in emitter looks (b+1)R from base;  R at base looks R/b from emitter.

CE unbypassed:  Zb = b re + (b+1)RE ~ b(re+RE);  Zi = R1||R2||Zb;  Zo = RC
                Av = -RC/(re+RE) ~ -RC/RE   (no beta -> stable), 180 deg
Partial bypass: DC uses RE1+RE2;  AC uses RE1 only:  Av = -RC/(re+RE1)
Emitter foll.:  Zi = RB||b(re+RE||RL)  (RL changes Zi!)
                Zo = RE||re (ideal source);  Zo = RE||(re + (Rs||RB)/b) with Rs
                Av = RE/(RE+re) ~ 1, AvL = (RE||RL)/(RE||RL + re), in phase
Common base:    Zi = RE||re ~ re (tiny);  Zo = RC;  Av = aRC/re (in phase);  Ai = -a
Coll. feedback: Av = -(RC||RF)/re  [= -(RF/(RC+RF))RC/re];  Zo = RC||RF
                Zi = re/(1/b + RC/(RC+RF))  =  b re || RF/(1-Av)  (Miller; use AvL i
Loading:        AvL = AvNL * RL/(RL+Zo)   (two-port; exact for CE types only)
                CE: replace RC by RC||RL.   Avs = Zi/(Zi+Rs) * AvL.   AiL = -AvL*Zi/R
                |AvNL| > |AvL| > |Avs|  always
ro: ignore only if ro >= 10RC (say so). Bypassed CE: Av = -(RC||ro)/re.
Prac measurement: Zi = Rs Vi/(Vs-Vi);  Zo = (VoNL - VoL)/VoL * RL
```

EDC · T7

Multistage amplifiers

Cheat sheet · Day 14 · Fri 6 Nov · pack E8

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
MULTISTAGE (T7)
Cascade: AvT = Av1*Av2*...  each Av computed with the NEXT stage's Zi as load
  Avs = Zi1/(Zi1+Rs) * AvT      AvL = AvNL * RL/(RL+Zo)    (two-port; Zo(CE)=RC)
  Two-port chain: Avs = [Zi1/(Zi1+Rs)] AvNL1 [Zi2/(Zi2+Zo1)] AvNL2 [RL/(RL+Zo2)]
Zi:  CE bypassed   R1||R2||b*re          CE unbypassed  R1||R2||b(re+RE)
     EF            RB||b(re + RE||RL)
Av:  CE bypassed  -(RC||Rload)/re        CE unbypassed  -(RC||Rload)/(re+RE) ~ -(RC|
     EF   (RE||RL)/(re+RE||RL) ~ 1       CB  +RC/re
Phase: CE inverts; CE-CE = +ve gain; EF, CB no inversion
CHECK: |Vo| = |AvT| Vi < available swing (~VCE)? else clipping
Lecture Wk5 ex (both bypassed, b=200, re=6.34): Av1=-104, Av2=-347, AvT=36.1e3; RL=4
Prac4 Fig1 (stage 2 UNBYPASSED): Zi2=3.52k, Av1=-212, Av2=-2.19 -> 463; RL=4.7k -> 3
Darlington: bD=b1*b2, VBE=2 drops (1.4 V; 1.6 V Boylestad)
  IB=(VCC-VBE)/(RB+bD*RE), Zi=RB||bD*RE, Ai~bD*RB/(RB+bD*RE), Av~1
Feedback pair: IB1=(VCC-VEB1)/(RB+b1*b2*RC), one VBE drop
Cascode: VB1=R3/(R1+R2+R3)*VCC, VB2=(R2+R3)/(R1+R2+R3)*VCC, IE=(VB1-0.7)/RE same both
  VC1=VB2-0.7, Av1=-1, Av2=+RC/re, Av~-RC/re; benefit = bandwidth (low Miller), not g
Direct-coupled: VB2 = VC1 = VCC - IC1*RC1 (check IB2 << IC1, else KCL at node)
Mirror: IREF=(VCC - VB1)/R, VB1 = emitter rail + 0.7;  Io = IREF/(1+2/b) -> IREF
  Design: R = (VCC - VB1)/[Io(1+2/b)]; keep output BJT active (VCE>=0.2)
npn: VE = VB - 0.7     pnp: VE = VB + 0.7     beta = inf -> IB = 0
```

FET physics

Cheat sheet · Day 15 · Sat 7 Nov · pack E9

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
FET PHYSICS (Week 6)                          I_G = 0  ->  I_S = I_D
JFET / D-MOSFET (Shockley, saturation):
  I_D = I_DSS (1 - VGS/VP)^2       VGS = VP (1 - sqrt(I_D/I_DSS))
  I_DSS: VGS = 0, VDS >= |VP|  (max for JFET; D-MOSFET can exceed with VGS>0)
  JFET: I_D = 0 for VGS beyond VP (VP = VGS(off))
  Sketch: (0, I_DSS) (0.3VP, I_DSS/2) (VP/2, I_DSS/4) (VP, 0)  concave up
  Ohmic: r_d = r_o / (1 - VGS/VP)^2     Knee: VDS(sat) = VGS - VP
E-MOSFET:
  I_D = k (VGS - VT)^2  for VGS > VT (n), else 0      no I_DSS!
  k = I_D(on) / (VGS(on) - VGS(Th))^2   [mA/V^2]
  VGS = VT +/- sqrt(I_D/k): keep the root that turns it ON
  VDS(sat) = VGS - VT
p-CHANNEL: same equations, signed values.
  p-JFET, p-DMOS: VP > 0.  p-EMOS: VT < 0, on for VGS < VT.  VDS < 0.
OUTPUT char = I_D vs VDS (VGS steps).  TRANSFER = I_D vs VGS (in saturation).
  Output -> transfer: read I_D at each VGS at one saturation VDS.
INVERTER / CMOS:
  NM_L = V_IL - V_OL     NM_H = V_OH - V_IH
  Ideal: V_OH = VDD, V_OL = 0, V_IL = V_IH = VDD/2 -> NM = VDD/2
  CMOS: VGS,n = Vi ; VGS,p = Vi - VDD.  Vi=0 -> pMOS on, Vo=VDD; Vi=VDD -> nMOS on, V
  One device always off -> ~no static power
  t_P = (t_PLH + t_PHL)/2, each measured to the output crossing VDD/2
```

FET DC bias

Cheat sheet · Day 17 · Mon 9 Nov · pack E10

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
FET DC BIAS  (I_G = 0, I_D = I_S, no drop across R_G; work in mA and kOhm)
JFET/D-MOS:  I_D = IDSS(1 - VGS/VP)^2      E-MOS: I_D = k(VGS - VT)^2,
             VGS = VP(1 - sqrt(ID/IDSS))          k = ID(on)/(VGS(on) - VT)^2
Sketch: VGS = 0, 0.3VP, 0.5VP, VP  ->  ID = IDSS, IDSS/2, IDSS/4, 0
Bias lines:  fixed   VGS = -VGG
             self    VGS = -ID*RS
             divider VGS = VG - ID*RS,  VG = R2*VDD/(R1+R2)
             CG/-VSS VGS = VSS - ID*RS   (VS = -VSS + ID*RS)
             E-MOS drain feedback  VGS = VDS = VDD - ID*RD
Output:  VD = VDD - ID*RD,  VDS = VD - VS
Method:  bias line -> sub into device eqn -> quadratic -> keep root with
         VP <= VGS <= 0 (JFET), VGS >= VP (D-MOS, can be +), VGS > VT (E-MOS)
Design:  target VD -> ID = (VDD-VD)/RD;  ID -> VGS (invert);
         VS = VG - VGS;  RS = VS/ID;  RD = (VDD - VD)/ID
ALWAYS CHECK SATURATION: VDS >= VGS - VP  (E-MOS: VDS >= VGS - VT)
  Fails (or VDS < 0) -> ohmic region, Shockley Q-point not valid; say so.
```

FET small-signal I

Cheat sheet · Day 18 · Tue 10 Nov · pack E11

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
FET SMALL-SIGNAL (Week 8)          ALWAYS: DC Q-point -> gm, rd -> ac circuit
gm0 = 2 IDSS/|VP|        gm = gm0(1 - VGSQ/VP) = gm0*sqrt(IDQ/IDSS)   (JFET, D-MOS)
E-MOS: gm = 2k(VGSQ - VGS(Th))     gm = gfs = yfs      rd = 1/gos = 1/yos
Model: G-S open | gm*Vgs (D->S) || rd | same model for JFET/D-MOS/E-MOS
Device only: Zo = rd, Av(FET) = -gm*rd
CS fixed / self-bias(Cs) / divider(Cs):  Zi = RG or R1||R2,  Zo = RD||rd,
                                         Av = -gm(RD||rd)  (~ -gm RD if rd >= 10RD)
CS unbypassed RS (rd ignored): Zi = RG, Zo ~ RD, Av = -gm RD/(1 + gm RS)
   with rd: Av = -gm RD/(1 + gm RS + (RD+RS)/rd)
Load/source: Av = -gm(rd||RD||RL); Avs = Zi/(Zi+Rsig) * Av; Zo excludes RL
DC reminders: fixed VGS=-VGG; self VGS=-ID RS; divider VGS=VG-ID RS;
   D-MOS with G,S at 0 V -> ID = IDSS, gm = gm0.  Pick root with VGS > VP (or > VGS(T
   Saturation check: VDS > VGS - VP  (E-MOS: VDS > VGS - VGS(Th))
Design for gain: RD||rd = |Av|/gm -> RD = 1/(1/(RD||rd) - 1/rd); recheck VDS
Graphical gm: tangent slope at Q, or chord symmetric about Q (exact for parabola)
Link to BJT: 1/gm plays the role of re.  CS output 180 deg out of phase.
```

FET small-signal II

Cheat sheet · Day 20 · Thu 12 Nov · pack E12

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
FET SMALL-SIGNAL II (Week 8 second half)
gm: JFET/D-MOS gm0 = 2IDSS/|VP|, gm = gm0(1 - VGSQ/VP)
    E-MOS gm = 2k(VGSQ - VGS(Th)),  k = ID(on)/(VGS(on)-VGS(Th))^2
    Any:  gm = 2ID/|VGS - Voff|      rd = 1/gos  (not given -> infinity)
Sedra: Vov = VGS - Vt, ID = 1/2 unCox (W/L) Vov^2, gm = unCox(W/L)Vov = 2ID/Vov
    course k = 1/2 unCox (W/L).  If W/L is asked, given "k" = unCox.
CONFIG        Zi                        Zo               Av
CS (byp/VD)   RG or R1||R2              RD||rd           -gm(RD||rd)
CS unbyp RS   RG                        ~RD              -gmRD/(1+gmRS+(RD+RS)/rd)
Src follower  RG                        rd||RS||1/gm     gm(rd||RS)/(1+gm(rd||RS))  <1
Common gate   RS||(rd+RD)/(1+gm rd)     RD||rd           (gmRD+RD/rd)/(1+RD/rd) ~ gmRD
              ~ RS||1/gm
Drain fdbk    (RF+rd||RD)/(1+gm(rd||RD)) RF||rd||RD     -gm(RF||rd||RD) ~ -gmRD
              ~ RF/(1+gmRD)
LOADING: put RL in parallel with output resistor (RD, or RS for follower).
         Avs = Zi/(Zi+Rsig) * AvL   (big effect only for CG)
CASCADE: AvT = Av1*Av2, Zi = Zi1, Zo = Zo(last), RL1 = Zi2; Vo = AvT*Vi.
         Don't round stage gains before multiplying.
DESIGN:  RD||rd = |Av|/gm -> RD = 1/(gm/|Av| - 1/rd); standard value; recheck
         VDS > VGS - VP (saturation).
Sedra quick: CG matched -> 1/gm = Rsig, Gv = 1/2 gmRD; follower Ro = 1/gm,
         Av = RL/(RL+1/gm); ID = gm Vov/2; W/L = gm/(unCox Vov).
```

Op-amps, differential amps, power amps

Cheat sheet · Day 21 · Fri 13 Nov · pack E13

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
DIFF / CMRR
 Vd = Vi1 - Vi2     Vc = (Vi1 + Vi2)/2     Vo = Ad*Vd + Ac*Vc
 CMRR = Ad/Ac       CMRR(dB) = 20 log10(Ad/Ac)    ratio = 10^(dB/20)
 Vo = Ad*Vd*(1 + Vc/(CMRR*Vd))          (use the ratio, not dB)
 Measure: Ad = Vo/Vd (Vc=0);  Ac = Vo/Vc (inputs tied)
BJT DIFF PAIR
 IE(tail) = (VEE - 0.7)/RE ; IC = IE/2 ; VC = VCC - IC*RC ; re = 26mV/IC
 single-ended Av = RC/(2re) ; double-ended Ad = RC/re ; Ac ~ RC/(2RE)
 Current mirror: IX = (Vsupply - VBE)/RX, I_out ~ IX
IDEAL OP-AMP: no input current; Vplus = Vminus (neg. feedback)
 Inverting  Vo = -(Rf/R1)V1         Non-inv  Vo = (1 + Rf/R1)V1
 Follower   Vo = V1                 Summer   Vo = -Rf*sum(Vk/Rk)
 Cascade: multiply stage gains, keep signs; check |Vo| < rails
POWER: eta = Po(ac)/Pi(dc); PQ = Pi - Po;  P = Irms^2 R = Ip^2 R/2 = Ipp^2 R/8
 Class A series: IB=(VCC-0.7)/RB, ICQ=bIB, VCEQ=VCC-ICQ*RC
   Pi = VCC*ICQ ; Po = Ic(rms)^2*RC ; Ic,pk(max) = min(ICQ, VCC/RC - ICQ)
   centred: Po,max = VCC^2/(8RC), eta max 25%
 Class A transformer: RL' = a^2 RL, a = sqrt(RL'/RL) ; eta max 50%
   Po = (VCEmax-VCEmin)(ICmax-ICmin)/8 ; Pi = VCC*ICQ
 Class B: Ip = VL(p)/RL ; Pi = 2VCC*Ip/pi ; Po = VL(p)^2/(2RL)
   eta = (pi/4)(VL(p)/VCC), max 78.5% ; P2Q = Pi - Po (both), PQ = P2Q/2
   max: Po = VCC^2/(2RL), Pi = 2VCC^2/(pi RL)
   P2Q,max = 2VCC^2/(pi^2 RL) at VL(p) = 0.636 VCC
```